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Alizée Pathology, LLC Tissue Cross Reactivity Report

Alizée Study Number DGG14-010

**Study
Title** Full Human Tissue Cross-Reactivity
Study of LT3114

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Date 1/13/2015

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1.0 Compliance Statement

All portions of this study performed at the testing facility adhered to the study protocol and Alizée Pathology's standard operating procedures (SOPs), and, except as noted below, the conduct of these portions of the study conformed to the most recent version of the following Good Laboratory Practice (GLP) standards: The Nonclinical Laboratory Studies Good Laboratory Practice Regulations issued by the U.S. Food and Drug Administration (Title 21 of the Code of Federal Regulations, Part 58).

Exceptions to the GLP compliant processes included the following:

- The optimization and validation of test article concentrations as well as other variable parameters of the immunohistochemistry methodology conducted under Alizée Pathology Study DGG14-010 (Pilot) performed by the testing facility (Alizée Pathology) constitutes unaudited method development and was conducted in accordance with Alizée Pathology SOPs, appropriate scientific methodology, and sound scientific judgment.
- Analysis of test and control article concentration was not conducted as specified under 21 CFR, Part 58.113 (Mixtures of Articles and Carriers). Initial concentration of test article and isotype control article is established by the Sponsor, as detailed below, and direct testing of final concentration(s) in carrier is impractical due to the admixture of other proteins and immunoglobulins. Mixture and dilution of the test article and isotype control article was conducted according to relevant Alizée SOPs and study documentation and was subject to quality assurance audit. Empirical validation of test article reactivity at final working dilutions was verified with positive and negative control samples in each staining run. Retention samples of test article and isotype control article and all documentation necessary to reconstruct working dilutions was retained for any necessary further analysis. Documentation of test and isotype control articles and dilutions used was also subject to quality assurance audit.

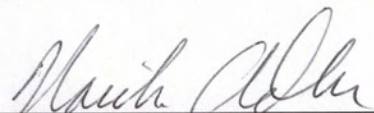
An exception to the study protocol was:

- It is noted that the lot of Sigma Fast 3,3' Diaminobenzidine (DAB) Solution (*Sigma-Aldrich*) used in this study, lot number SLBH5441V, was not the lot specified in the study protocol, section XII. Immunoperoxidase Staining, C. Reagents. The lot specified in the protocol was lot number SLBJ3609V.

In the opinion of the Study Director, these exceptions to GLP compliance did not impact the quality, integrity, or validity of this study. Additional exceptions to full GLP compliance, if any, have been documented in the raw data and in the compliance statement of the final report.

Compliance of any portion of the study performed by the Sponsor, such as test and control article characterization including appropriate testing for identity, stability, strength, purity, composition or other characteristics for each batch or lot and methods of synthesis, fabrication, or derivation of the test and control articles and stability testing as applicable is the responsibility of the Sponsor.

Study Director:



Naïche L. Adler, PhD

1-13-2015
Date

2.0 Executive Summary

The purpose of this study was to evaluate potential tissue cross-reactivity of LT3114-CF on cryosections of normal human tissues.

Methods: LT3114-CF (test article) and LT1023-CF isotype control were used to stain cryosections from 38 normal human tissues, from at least three separate donors (except as noted in 8.6 Test Tissues), at two concentrations (1 and 10 µg/mL), using an indirect immunohistochemistry procedure. Antibody dilution buffer without primary antibody was used as an assay control.

Conclusions: Using an indirect immunoperoxidase staining procedure, LT3114-CF was applied at 1 and 10 µg/mL to a full panel of human tissues based on recommendations in Annex II of Directive 75/318/EEC, "Production and Quality Control of Monoclonal Antibodies" and in the 1997 US FDA/CBER "Points to Consider in the Manufacture and Testing of Monoclonal Antibody Products for Human Use." There was no LT3114-CF specific staining, and therefore no unexpected reactivity, in any of the tissues evaluated.

3.0 Introduction

3.1 Protocol

This report presents and compares the results of immunohistochemical evaluation of the reactivity of the test article, LT3114-CF, and the isotype control article, LT1023-CF. LT3114-CF (test article) is a humanized monoclonal IgG1 (κ) antibody directed against lysophosphatidic acid (LPA) and has been conjugated to 5-(6)-carboxyfluorescein (CF), supplied at a concentration of 3.6 mg/mL, at pH 7.4. The isotype control antibody, LT1023-CF, is a humanized non-specific monoclonal IgG1 (κ) antibody and has been conjugated to 5-(6)-carboxyfluorescein (CF), supplied at a concentration of 2.9 mg/mL, pH 7.4. Joan R. Wicks, DVM, PhD, DACVP served as the Study Pathologist. Naiche L. Adler, PhD served as the Study Director and Peer Review Scientist.

Method development to determine the reagent concentrations and staining conditions to be employed in the tissue cross-reactivity study was conducted under study number DGG14-010 (Pilot), in accordance with Alizée Standard Operating Procedures (SOPs) and sound scientific judgment, although quality assurance inspections were neither required nor performed. The reagent concentrations determined in optimization staining runs were validated by reproducibility of the positive controls in this tissue cross-reactivity study.

3.2 Purpose

The purpose of this study was to evaluate potential tissue cross-reactivity of LT3114-CF on cryosections of normal human tissues.

4.0 Study Timetable

- Study Initiation Date: November 18, 2014
- Study Completion Date: date of signature of report by Study Director (see section 14.0 Signatures).

5.0 Test and Isotype Control Articles

5.1 Test Article

Description: LT3114-CF (test article) is a humanized monoclonal IgG1 (κ) antibody directed against lysophosphatidic acid (LPA) and has been conjugated to 5-(6)-carboxyfluorescein (CF), supplied at a concentration of 3.6 mg/mL, pH 7.4.

Identification Number (Lot): FAM999-14-005.

Manufacturer: The unconjugated test antibody (LT3114) was manufactured by Gallus-MO on behalf of Lpath Inc. CF conjugation was performed by Columbia Biosciences. The final concentration of the solution was documented in the raw data.

Storage Conditions: Stored at 2-8°C.

Stability: Stability of LT3114-CF is unknown. Information on stability under conditions of administration (room temperature) is included in this report if it has been determined. Stability testing of the test antibody (LT3114) was conducted for 3 months at 5±3°C. Results indicated that the lot tested (lot 103948) conformed to the reference standard at 3 months under this storage condition.

Test Article Characteristics: Information on identity, purity, and composition, or other characteristics that define the Test Article are on file with the manufacturer. A Certificate of Analysis has been provided and the lot/batch of the test article was retained in the working folder and included in the final report.

Handling Precautions: A Material Safety Data Sheet (MSDS) is on file in the working folder for this study.

5.2 Isotype Control Article

Description: LT1023-CF (isotype control article), is a humanized non-specific monoclonal IgG1 (κ) antibody and has been conjugated to 5-(6)-carboxyfluorescein (CF), supplied at a concentration of 2.9 mg/mL, pH 7.4.

Identification Number: Lot number FAM999-14-006.

Manufacturer: The unconjugated isotype control antibody (LT1023) was manufactured and supplied by Lpath, Inc. CF conjugation was performed by Columbia Biosciences. The final concentration of the solution was documented in the raw data.

Storage Conditions: Store at 2 to 8°C.

Stability: Stability of LT1023-FC and LT1023 is unknown. Information on stability under conditions of administration (room temperature) has not been determined.

Control Article Characteristics: Information on identity, purity, stability, and composition, or other characteristics that define the control article are on file with the manufacturer. The lot of the control article was recorded in the working folder and included in this report (see Identification Number (Lot) above).

Handling Precautions: An MSDS is on file in the working folder for this study.

6.0 Contributing Personnel

<u>Testing Facility Role:</u>	<u>Personnel:</u>	<u>Address:</u>
Study Director	Naiche L. Adler, PhD	Alizée Pathology
Study Pathologist	Joan R. Wicks DVM, PhD, DACVP	20 Frederick Road
Peer Review Scientist	Naiche L. Adler, PhD	Thurmont, MD 21788
Testing Facility Management	Kathy Bailey	
Senior Laboratory Specialist	Randy Hart, BS	
<u>Sponsor Role:</u>	<u>Personnel:</u>	<u>Address:</u>
Sponsor Representative	Gary Woodnutt, PhD	Lpath, Inc. 4025 Sorrento Valley Blvd. San Diego, CA 92121
Study Monitor	Marlene W. Modi, PhD	MWM Consulting Group, Inc. for Lpath, Inc. 14 Joseph Court Monmouth Junction, NJ 08852
<u>Archives Facility Role:</u>	<u>Personnel:</u>	<u>Address:</u>
Document Center & Archive Manager	Terri L Gyza	Calvert Laboratories, Inc. Scott Technology Park 130 Discovery Drive Scott Township, PA 18447

7.0 Material Disposition

Test article accountability was maintained according to Alizée Pathology SOPs. Concurrent with finalization of the study, the remaining test and isotype control article will be reconciled and disposition will be based on direction from the Sponsor. The retention samples of the test article, the isotype control, and all other items listed under XIII. Records in the study protocol will be stored and archived at Calvert Laboratories, Inc., Scott Technology Park, 130 Discovery Drive, Scott Township, PA 18447, with Terri L. Gyza, Document Center & Archive Manager, serving as archivist.

8.0 Test System (Cryosections of Human Tissues)

8.1 Source

Tissues were obtained at autopsy or biopsy from humans. Pertinent data (*i.e.*, identification, source, and age) for tissues used on study are on file at Alizée Pathology. Tissues as received were considered to be essentially normal based upon clinical history and review by Alizée pathologists. Each human donor was assigned a unique identification number that was documented in the working folder.

8.2 Justification for Selection of Test System

Tissue cross-reactivity studies are conducted as part of the investigational new drug (IND) submission for therapeutic antibodies and other biologics. The purpose of this study was to evaluate the potential reactivity of LT3114-CF on cryosections of normal human tissues. The test system had no requirements for bodyweight range, sex, or age of donors other than for selection of normal tissues and sex-specific tissues.

8.3 Test System Identification

Upon receipt at Alizée, donor tissue was identified with a unique permanent number which was recorded in the appropriate Master Tissue Bank and Individual Donor Inventory Sheet forms, on all relevant paperwork, and on the protective sealable and freezer-safe container. This number and any associated sub-numbers were used as references throughout the study.

8.4 Tissue Embedding and Storage

Unfixed tissue samples were placed in molds and frozen on dry ice in optimal cutting temperature (OCT) compound embedding medium. Tissues were placed in sealed plastic bags and maintained below -70°C until sectioning.

8.5 Tissue Sectioning

Tissue sectioning was performed according to the Alizée Pathology cryotomy SOP. Sections were cut at approximately $4\text{-}5\text{ }\mu\text{m}$ and post-fixed for 10 minutes in acetone followed by 10 seconds in 10% neutral buffered formalin (NBF).

8.6 Test Tissues

Specimens of the following tissues from three different human donors were stained and evaluated, except where noted:

Adrenal	Kidney (glomerulus, tubule)	Skin
Blood cells*	Liver	Spinal cord
Blood vessels (endothelium)**	Lung	Spleen
Bone marrow	Lymph node	Striated (skeletal) muscle
Brain – cerebrum (cortex)	Ovary	Testis
Brain - cerebellum	Fallopian tube (oviduct)	Thymus
Breast (mammary gland)***	Pancreas	Thyroid
Eye	Parathyroid****	Tonsil
Gastrointestinal tract – colon (large intestine)	Peripheral nerve	Ureter
Gastrointestinal tract – esophagus	Pituitary	Urinary bladder
Gastrointestinal tract – small intestine	Placenta	Uterus – body (endometrium)
Gastrointestinal tract – stomach	Prostate	Uterus – cervix
Heart	Salivary (parotid) gland	

* Blood cells include granulocytes, lymphocytes, monocytes and platelets and were evaluated on blood smears.

** Endothelium evaluated in all tissues except blood smears.

*** Despite multiple attempts, only two donors were available for evaluation of breast (mammary gland).

**** Despite multiple attempts, only two parathyroid donors were available for evaluation.

9.0 Test System (Cryosections of Human Tissues) Immunoperoxidase Staining

9.1 Rationale for Test Article Concentration Selection

Staining validation was conducted under study number DGG14-010 (Pilot) and is summarized in Appendix A to the Study Protocol, DGG14-010 (Pilot) Executive Summary Immunohistochemistry Report, included in this report in Appendix C. Study Protocol and Amendment. Using an indirect immunoperoxidase technique, the ideal concentration for the test article and the isotype control article determined for the study (*i.e.*, the lowest concentration of test article showing no reduction in staining intensity or frequency) was 1 µg/mL. The higher concentration of test article determined for the human cross-reactivity study was a 10-fold increase over the ideal concentration, *i.e.* 10 µg/mL. Thus, the staining concentrations for the test article and isotype control article used for the human tissue cross-reactivity study were 1 and 10 µg/mL. The staining procedure was the same as that used in method development study DGG14-010 (Pilot).

9.2 Working Antibody Concentrations for Testing

The working concentrations of the test article and isotype control article used in this study were: 1 µg/mL and 10 µg/mL.

9.3 Method Controls

- **Positive Control:** Positive control material, LPA(C12)-SMCC-BSA, is a 12-carbon lysophosphatidic acid (LPA) molecule conjugated to bovine serum albumin (BSA) and immobilized by spotting to a glass microscope slide (spot slide). Spot slides were prepared by diluting 0.05 µg of LPA(C12)-SMCC-BSA in carbonate buffer (100 mM NaHCO₃, 33.6 Na₂CO₃, at pH 9.5) and applying 10 µL of that mixture to glass microscope slides. The mixture was dried on the slide at 37°C for 30 minutes.
- **Negative Control:** Negative control material, S1P-SMCC-BSA, is sphingosine-1-phosphate (S1P), also conjugated to BSA and immobilized by spotting to a glass microscope slide (spot slide). S1P has been demonstrated by the Sponsor to not react with the unconjugated version of the test article (LT3114) *in vitro*. Spot slides were prepared by diluting 0.05 µg of S1P-SMCC-BSA in carbonate buffer, and applying 10 µL of that mixture to glass microscope slides. The mixture was dried on the slide at 37°C for 30 minutes.

- Isotype Control Article: LT1023-CF.
- Assay Control: The assay control acted as the control for all aspects of the staining procedure aside from the primary antibody (test article or isotype control article). The immunoperoxidase staining method was identical to that used for the test article except that dilution buffer was used as a replacement for the primary antibody.
- Tissue Staining Control: The adequacy of all tissues used in this study (with the exception of the positive and negative control tissues and the blood smear) was validated through immunohistochemical staining for the presence of vimentin. Vimentin is an intracellular structural protein expressed by many cell types and was present in all tissues used for this study.

9.4 Reagents

Reagents for Test Article and Isotype Control Article Staining

- Biocare Tris-buffered saline Auto-Wash (TBS-AW) (*Biocare Medical*)
- Peroxidized® 1 (*Biocare Medical*)
- Peroxidized® 1 Diluent (*Biocare Medical*)
- Human IgG (HulgG) (*Sigma-Aldrich, catalog number I4506, lot number 121M7009V*)
- Human γ -globulins (HGG) (*Sigma-Aldrich, catalog number G4386, lot number SLBD8893V*)
- Avidin/Biotin blocking kit (*Vector Laboratories*)
- Mouse monoclonal anti FITC (*Sigma Aldrich, catalog number F5636, lot number 111M4794*)
- 14974 antibody diluent (provided by Calvert Laboratories, Inc.)
- Dako EnVision+ System–HRP (peroxidase polymer, anti-mouse) (*Dako, catalog number K4007, lot number 10088650*)
- Sigma Fast 3,3' Diaminobenzidine (DAB) Solution (*Sigma-Aldrich, catalog number D4414, lot number SLBH5441V*)
- Neutral Buffered Formalin (NBF)
- Acetone (*Mercedes Medical*)

Reagents for Staining with Anti-Vimentin Antibody

- Avidin/Biotin Blocking Kit (*Vector Laboratories*)
- Peroxidized® 1 (*Biocare Medical*)
- Peroxidized® 1 Diluent (*Biocare Medical*)
- Sniper® (*Biocare Medical*)
- Anti-human vimentin antibody (*Biocare Medical*)
- Goat anti-Mouse IgG – Biotinylated (*Jackson*)
- ABC Elite (*Vector Laboratories*)
- Sigma Fast DAB (*Sigma-Aldrich*)
- DaVinci Green Diluent (*Biocare Medical*)
- Biocare TBS Auto-wash (*Biocare Medical*)
- 10% Neutral Buffered Formalin (NBF)
- Acetone (*Mercedes Medical*)

All reagents used were documented in the working folder (using Alizée Form 825-2, Frozen Immunohistochemistry Staining Run Sheet) for this study.

9.5 Reaction Sequence: Test and Isotype Control Antibodies

An indirect immunoperoxidase procedure was performed. Primary and secondary diluent for the test article and isotype control article antibodies reaction sequence was 14974 (24.3 mM citrate, 51.4 mM sodium phosphate, 300 mM D-trehalose pH 5.0 containing 2 mL 10% polysorbate 80), provided to Alizée by Calvert Laboratories, Inc. Unless otherwise indicated, all rinses were performed using TBS-AW and all steps were performed at room temperature.

Prior to staining, slides were fixed in cold acetone for 10 minutes, allowed to air dry, and then fixed in 10% NBF for 10 seconds followed by a two-minute rinse in distilled water, and two rinses with TBS-AW.

Avidin complex was applied for 15 minutes, and the slides were rinsed once. A Biotin complex was applied for 15 minutes, followed by one rinse. Next, 1 mg/mL heat aggregated human IgG in PBS, with 0.5% casein and 5% human gamma globulins was applied as a protein block for 30 minutes. The slides were then incubated in primary antibody solution (or dilution buffer for the assay control) for 60 minutes and rinsed twice. Peroxidized I (diluted 1:5 in Peroxidized I Diluent) was applied for 10 minutes to reduce endogenous peroxidase staining and rinsed twice. Mouse anti-FITC IgG was applied for 30 minutes, and after two rinses, Dako Envision+ anti-mouse IgG HRP polymer was applied for an additional 30 minutes. Color development was performed by incubation for 5 minutes with Sigma Fast DAB solution, followed by a rinse with distilled water. The slides were counterstained and coverslipped according to Alizée SOPs.

9.6 Tissue staining control (Validation of Tissue Samples using Anti-Vimentin Staining): Reaction Sequence

Unless otherwise indicated, rinses were performed using TBS-AW, and all steps were performed at room temperature. DaVinci Green Diluent served as the diluent for primary and secondary antibodies. Prior to staining, slides were fixed in cold acetone for 10 minutes, allowed to air dry, and then fixed in 10% NBF for 10 seconds followed by a two-minute rinse in distilled water. The slides were then rinsed two times. An Avidin block solution was applied for 15 minutes, then rinsed once. Next a Biotin block solution was applied for 15 minutes, then rinsed once. A block to reduce endogenous peroxidase staining using Peroxidized I (diluted 1:5 in Peroxidized I Diluent) was applied for 5 minutes, then rinsed two times. Sniper protein block was applied for 20 minutes to reduce non-specific binding, then rinsed two times. Primary antibody solution, anti-vimentin antibody and DaVinci Green antibody diluent at a 1:500 ratio, was applied and allowed to incubate for 60 minutes. Slides were rinsed twice. A biotinylated anti-mouse IgG secondary antibody at a 1:500 ratio was applied for 30 minutes. Tissues were rinsed two times. ABC Elite was applied for 30 minutes followed by two rinses. Color development was performed by incubation for 5 minutes with Sigma Fast DAB solution. The slides were rinsed in distilled water, and counterstained and coverslipped according to Alizée SOPs.

10.0 Evaluation

Except as noted in Appendix A. Tabulated Microscopic Data, all immunohistochemically stained slides, including positive and negative control slides and assay control slides, were evaluated by Joan R. Wicks, DVM, PhD, DACVP (Study Pathologist) and the adequacy of tissue morphology and staining was determined. Data reporting included identification of the tissue element, or cell type, showing reactivity, and frequency and average intensity of the staining, with elements that were negative for staining consolidated under 'Other Elements' for each tissue (if applicable). When appropriate, the subcellular localization of staining was also reported. The intensity of specific reactivity was graded as follows: 0 = negative; 1 = weak (blush); 2 = mild; 3 = moderate; 4 = strong; 5 = intense.

The frequency of specific reactivity was graded according to the percent of a particular cell type or tissue element as follows: N = negative, no reactivity; VR = very rare, less than 5% of cells of that cell type; R = rare, 5-25% of cells of that cell type; O = occasional, 26-50% of cells of that cell type; F = frequent, 51-75% of cells of that cell type; VF = very frequent, 76 – 100% of cells of that cell type. All data was recorded in a Microsoft® Excel spreadsheet and results are presented in Appendix A. Tabulated Microscopic Data.

The Alizée Pathology Peer Review Scientist, Naiche L. Adler, PhD, examined slides and reviewed the report. Evaluation data recorded by the Study Pathologist was provided for reference. Alizée's peer review statement is included in Appendix B. Peer Review Statement.

The study protocol and amendment are included in Appendix C. Study Protocol and Amendment. The certificates of analysis for the test article and isotype control article are included in Appendix D. Certificates of Analysis. A Quality Assurance Statement is included as Appendix E. Quality Assurance Statement.

Due to the semiquantitative nature of the data, statistical analysis was not performed.

11.0 Results

The results of tissue staining for specific donors and tissues are presented in Appendix A. Tabulated Microscopic Data. The tissues used in this study, with the exception of positive and negative control tissues and blood smears, were validated for use in immunohistochemistry through staining of serial or step cryosections using anti-vimentin antibody. The expression of vimentin was demonstrated in a variety of cell types (including endothelium) in all tissues.

Despite multiple attempts, only two parathyroid donors and two mammary gland donors were found within the time constraints of the study. The lack of a full contingent of donors for these two tissues weakens any conclusions specific to the tissues; however, this absence does not adversely affect the study as a whole.

11.1 Positive and Negative Controls

LT3114-CF produced intense staining of the positive control material, LPA[C12]-SMCC-BSA, at both 1 and 10 µg/mL. Neither antibody specifically reacted with the negative control material, S1P-SMCC-BSA. The isotype control antibody, LT1023-CF, did not specifically react with either the positive or negative control materials. There was no reactivity of the positive or negative control material when the test article and isotype control article were omitted from the staining reaction (assay control). Staining with anti-vimentin antibody validated the staining ability of the tissues evaluated in this study.

Staining of positive control material with the test article was inadequate in Runs 1 and 16, which precluded evaluation of the test tissues stained in those runs. Affected tissues were restained in subsequent runs and evaluated. See Appendix A. Tabulated Microscopic Data for detailed information on affected tissues and restaining.

11.2 Test Article Binding to Normal Human Tissues

LT3114-CF and the isotype control article, LT1023-CF were applied to a full panel of human tissues at 1 and 10 µg/mL. There was no LT3114-CF specific staining, and therefore no unexpected reactivity, in any of the tissues evaluated.

12.0 Conclusions

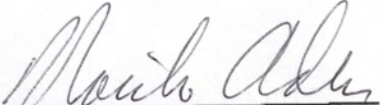
Using an indirect immunoperoxidase staining procedure, LT3114-CF was applied at 1 and 10 µg/mL to a full panel of human tissues based on recommendations in Annex II of Directive 75/318/EEC, "Production and Quality Control of Monoclonal Antibodies" and in the 1997 US FDA/CBER "Points to Consider in the Manufacture and Testing of Monoclonal Antibody Products for Human Use." There was no LT3114-CF specific staining, and therefore no unexpected reactivity, in any of the tissues evaluated.

13.0 References

1. Good Laboratory Practice for Nonclinical Laboratory Studies, Title 21, Code of Federal Regulations, Part 58, issued by the U.S. Food and Drug Administration (effective June 20, 1979)
2. "Points to Consider in the Manufacture and Testing of Monoclonal Antibody Products for Human Use," U.S. Department of Health and Human Services, Food and Drug Administration, Center for Biologics Evaluation and Research (CBER). February 28, 1997.
3. Annex II Directive 75/318/EEC, "Production and Quality Control of Monoclonal Antibodies," last revised December 1994.

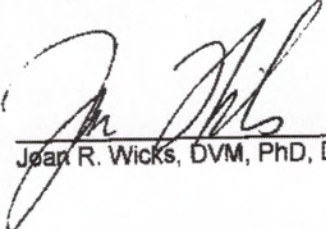
14.0 Signatures

Study Director:


Naïche L. Adler, PhD

1-13-2015
Date

Study Pathologist:


Jean R. Wicks, DVM, PhD, DACVP

1/13/15
Date

Appendix A. Tabulated Microscopic Data

DGG14-010	Scoring Definitions	Intensity of specific reactivity						Frequency of cells displaying specific reactivity				
		0	Negative						N	Negative, no reactivity		
		1	Weak (blush)						VR	Very rare, <5% of cells of that cell type		
		2	Mild						R	Rare, 5-25% of cells of that cell type		
		3	Moderate						O	Occasional, 26-50% of cells of that cell type		
		4	Strong						F	Frequent, 51-75% of cells of that cell type		
		5	Intense						VF	Very frequent, 76-100% of cells of that cell type		
	Relative Staining Strength by Location (tissue element or subcellular, when applicable):											
	* Due to inadequate staining of positive control material, tissues stained in Runs 1 and 16 could not be evaluated for test article reactivity and were restained in subsequent runs, with the exception of tissues which were unavailable or unsuitable for staining.											
Cryosection	Donor and Block	Run Number	Tissue Elements; Subcellular Localization	Specific Reactivity (Intensity/Frequency)					Assay Control	Tissue Comments	Vimentin	
				Test Article: LT3114-CF		Isotype Control Article: LT1023-CF		Run Number			Staining Results	
				10 µg/mL	1 µg/mL	10 µg/mL	1 µg/mL					
Positive Control Material (LPA(C12)-SMCC-BSA)	NA	*1, 3, 5, 7, 9, 11, 13, 15,*16, 18	NA	5	4-5	0	0	0	Staining of positive control material was inadequate in Runs 1 and 16. Therefore, the study tissue slides stained during those runs were not evaluated and were restained in subsequent runs.	NA	NA	
Negative Control Material (S1P-SMCC-BSA)	NA	*1, 3, 5, 7, 9, 11, 13, 15, *16, 18	NA	0	0	0	0	0	No comments.	NA	NA	
Adrenal	Hu37 AD-2	*1, 13	All elements	0/N	0/N	0/N	0/N	0/N	No comments.	2	Positive	
	Hu62 AD-2	*1, 13	All elements	0/N	0/N	0/N	0/N	0/N	No comments.	2	Positive	
	Hu103 AD-4	*1, 13, *16, 18	All elements	0/N	0/N	0/N	0/N	0/N	Tissue was missing in some slides in Run 13. Staining repeated in Runs 16 and 18.	2	Positive	
Blood Cells	Hu25 Blood Smear	15	All elements	0/N	0/N	0/N	0/N	0/N	Endogenous peroxidase (granulocytes) in all slides.	NA	NA	
	Hu160 Blood Smear	15	All elements	0/N	0/N	0/N	0/N	0/N	Endogenous peroxidase (granulocytes) in all slides.	NA	NA	
	Hu154 Blood Smear	15	All elements	0/N	0/N	0/N	0/N	0/N	Endogenous peroxidase (granulocytes) in all slides.	NA	NA	

DGG14-010	Scoring Definitions	Intensity of specific reactivity						Frequency of cells displaying specific reactivity				
		0	Negative						N	Negative, no reactivity		
		1	Weak (blush)						VR	Very rare, <5% of cells of that cell type		
		2	Mild						R	Rare, 5-25% of cells of that cell type		
		3	Moderate						O	Occasional, 26-50% of cells of that cell type		
		4	Strong						F	Frequent, 51-75% of cells of that cell type		
		5	Intense						VF	Very frequent, 76-100% of cells of that cell type		
		Relative Staining Strength by Location (tissue element or subcellular, when applicable):										
	* Due to inadequate staining of positive control material, tissues stained in Runs 1 and 16 could not be evaluated for test article reactivity and were restained in subsequent runs, with the exception of tissues which were unavailable or unsuitable for staining.											

Cryosection	Donor and Block	Run Number	Tissue Elements; Subcellular Localization	Specific Reactivity (Intensity/Frequency)					Tissue Comments	Vimentin	
				Test Article: LT3114-CF		Isotype Control Article: LT1023-CF		Assay Control		Run Number	Staining Results
				10 µg/mL	1 µg/mL	10 µg/mL	1 µg/mL				
Blood vessels (endothelium)	All tissues	*1,3, 5, 7, 9, 11, 13, 15,*16,18	NA	NA	NA	NA	NA	NA	Evaluated in all tissues which contain blood vessels. See individual tissues for findings.	2, 4, 6, 8, 10, 12,14,17	Positive
Bone Marrow	Hu100 BM-6	*1, 13	All elements	0/N	0/N	0/N	0/N	0/N	Endogenous peroxidase (myeloid elements) and endogenous pigment (hemosiderin). Staining of many cells throughout section in all slides attributed to binding by secondary or tertiary antibody.	2	Positive
	Hu105 BM-6	*1, 13	All elements	0/N	0/N	0/N	0/N	0/N	Endogenous peroxidase (myeloid elements) and endogenous pigment (hemosiderin). Staining of many cells throughout section in all slides attributed to binding by secondary or tertiary antibody.	2	Positive
	Hu108 BM-1	*1, 13	All elements	0/N	0/N	0/N	0/N	0/N	Endogenous peroxidase (myeloid elements) and endogenous pigment (hemosiderin). Staining of many cells throughout section in all slides attributed to binding by secondary or tertiary antibody.	2	Positive
Brain - cerebellum	Hu120 CL-1	*1, 15	All elements	0/N	0/N	0/N	0/N	0/N	Cytoplasmic staining of spindloid cells in meninges in all slides attributed to binding by secondary or tertiary antibody.	2	Positive
	Hu131 CL-3	*1, 15	All elements	0/N	0/N	0/N	0/N	0/N	Cytoplasmic staining of spindloid cells in meninges in all slides attributed to binding by secondary or tertiary antibody.	2	Positive
	Hu132 CL-4	*1, 15	All elements	0/N	0/N	0/N	0/N	0/N	Cytoplasmic staining of spindloid cells in meninges in all slides attributed to binding by secondary or tertiary antibody.	2	Positive
Brain - cerebrum (cortex)	Hu75 CR-1	*1, 13	All elements	0/N	0/N	0/N	0/N	0/N	Cytoplasmic staining of spindloid cells in meninges in all slides attributed to binding by secondary or tertiary antibody.	2	Positive
	Hu128 CR-1	*1, 13	All elements	0/N	0/N	0/N	0/N	0/N	Cytoplasmic staining of spindloid cells in meninges in all slides attributed to binding by secondary or tertiary antibody.	2	Positive
	Hu129 CR-2	*1, 13	All elements	0/N	0/N	0/N	0/N	0/N	No comments.	2	Positive

NA = Not applicable;
U=Unavailable.

DGG14-010	Scoring Definitions	Intensity of specific reactivity						Frequency of cells displaying specific reactivity			
		0	Negative					N	Negative, no reactivity		
		1	Weak (blush)					VR	Very rare, <5% of cells of that cell type		
		2	Mild					R	Rare, 5-25% of cells of that cell type		
		3	Moderate					O	Occasional, 26-50% of cells of that cell type		
		4	Strong					F	Frequent, 51-75% of cells of that cell type		
		5	Intense					VF	Very frequent, 76-100% of cells of that cell type		
		Relative Staining Strength by Location (tissue element or subcellular, when applicable):									
* Due to inadequate staining of positive control material, tissues stained in Runs 1 and 16 could not be evaluated for test article reactivity and were restained in subsequent runs, with the exception of tissues which were unavailable or unsuitable for staining.											
Cryosection	Donor and Block	Run Number	Tissue Elements; Subcellular Localization	Specific Reactivity (Intensity/Frequency)					Tissue Comments	Vimentin	
				Test Article: LT3114-CF		Isotype Control Article: LT1023-CF		Assay Control		Run Number	Staining Results
				10 µg/mL	1 µg/mL	10 µg/mL	1 µg/mL				
Breast (mammary gland)	Hu56 MG-3	*1, 15	U	U	U	U	U	U	No gland or duct tissue present. Replaced with alternate block (Hu56 MG-2).	2	Positive
	Hu56 MG-2	*16, 18	All elements	0/N	0/N	0/N	0/N	0/N	Replacement block for Hu56 MG-3. Non-specific staining of milk in all slides.	17	Positive
	Hu98 MG-1	*1, 15	All elements	0/N	0/N	0/N	0/N	0/N	No comments.	2	Positive
	Hu104 MG-1	*1, 15	U	U	U	U	U	U	No gland or duct tissue present. Attempted to replace with alternate block (Hu104 MG-2).	2	Positive
	Hu104 MG-2	*16, 18	U	U	U	U	U	U	Replacement block for Hu104 MG-1. No gland or duct tissue present. Tissue not used.	U	U
Eye	Hu141 EY-1	*1, 15	All elements	0/N	0/N	0/N	0/N	0/N	Choroid, iris, and ciliary body present. Endogenous pigment (melanin).	2	Positive
	Hu143 EY-2	*1, 15	All elements	0/N	0/N	0/N	0/N	0/N	Choroid, iris, and ciliary body present. Endogenous pigment (melanin).	2	Positive
	Hu150 EY-6	*1, 15	All elements	0/N	0/N	0/N	0/N	0/N	Lens, retina, sclera, choroid, ciliary body, conjunctiva and lens present. Endogenous pigment (melanin).	2	Positive
Fallopian Tube (oviduct)	Hu45 FT-3	7	U	U	U	U	U	U	Sample inadequate. Replaced with oviduct present on Hu108 OV-5	8	Positive
	Hu98 FT-1	7	U	U	U	U	U	U	Sample inadequate. Replaced with oviduct present on Hu98 OV-1	8	Positive
	Hu27 FT-1	7	All elements	0/N	0/N	0/N	0/N	0/N	Staining of occasional to frequent epithelial cells present in all slides attributed to binding by secondary or tertiary antibody.	8	Positive
	Hu98 OV-1	7	All elements	0/N	0/N	0/N	0/N	0/N	Replacement block for Hu98 FT-1. Fallopian tube present on slide containing ovary.	8	Positive
	Hu108 OV-5	7	All elements	0/N	0/N	0/N	0/N	0/N	Replacement block for Hu45 FT-3. Fallopian tube present on slide containing ovary.	8	Positive
Gastrointestinal tract - colon (large intestine)	Hu91 CO-4	3	All elements	0/N	0/N	0/N	0/N	0/N	Endogenous peroxidase (granulocytes). Nonspecific staining of mucus and fecal material, spindloid cells and smooth muscle cytoplasm variably present in all slides attributed to secondary antibody.	4	Positive
	Hu92 CO-4	3	All elements	0/N	0/N	0/N	0/N	0/N	Endogenous peroxidase (granulocytes). Nonspecific staining of mucus and fecal material, and spindloid cells variably present in all slides attributed to secondary antibody.	4	Positive
	Hu97 CO-3	3	All elements	0/N	0/N	0/N	0/N	0/N	Endogenous peroxidase (granulocytes). Nonspecific staining of mucus and fecal material, and spindloid cells variably present in all slides attributed to secondary antibody.	4	Positive

NA = Not applicable;
U=Unavailable.

DGG14-010	Scoring Definitions	Intensity of specific reactivity						Frequency of cells displaying specific reactivity				
		0	Negative						N	Negative, no reactivity		
		1	Weak (blush)						VR	Very rare, <5% of cells of that cell type		
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		3	Moderate						O	Occasional, 26-50% of cells of that cell type		
		4	Strong						F	Frequent, 51-75% of cells of that cell type		
		5	Intense						VF	Very frequent, 76-100% of cells of that cell type		
		Relative Staining Strength by Location (tissue element or subcellular, when applicable):										
	* Due to inadequate staining of positive control material, tissues stained in Runs 1 and 16 could not be evaluated for test article reactivity and were restained in subsequent runs, with the exception of tissues which were unavailable or unsuitable for staining.											
Cryosection	Donor and Block	Run Number	Tissue Elements; Subcellular Localization	Specific Reactivity (Intensity/Frequency)					Tissue Comments	Vimentin		
				Test Article: LT3114-CF		Isotype Control Article: LT1023-CF		Assay Control		Run Number	Staining Results	
				10 µg/mL	1 µg/mL	10 µg/mL	1 µg/mL					
Gastrointestinal tract - esophagus	Hu86 ES-2	3	All elements	0/N	0/N	0/N	0/N	0/N	Endogenous peroxidase (granulocytes). Nonspecific smudgy staining of several cell types variably present in all slides, but most prominent in assay control and isotype control article stained slides.	4	Positive	
	Hu90 ES-2	3	All elements	0/N	0/N	0/N	0/N	0/N	Endogenous peroxidase (granulocytes).	4	Positive	
	Hu103 ES-3	3	All elements	0/N	0/N	0/N	0/N	0/N	Endogenous peroxidase (granulocytes).	4	Positive	
Gastrointestinal tract - small intestine	Hu100 SI-1	3	All elements	0/N	0/N	0/N	0/N	0/N	Staining of endogenous peroxidase (granulocytes), and nonspecific staining of fecal material and mucus in all slides.	4	Positive	
	Hu99 SI-4	3	All elements	0/N	0/N	0/N	0/N	0/N	Staining of endogenous peroxidase (granulocytes), and nonspecific staining of fecal material and mucus in all slides.	4	Positive	
	Hu98 SI-1	3	All elements	0/N	0/N	0/N	0/N	0/N	Staining of endogenous peroxidase (granulocytes), and nonspecific staining of fecal material and mucus in all slides.	4	Positive	
Gastrointestinal tract - stomach	Hu41 ST-10	5	All elements	0/N	0/N	0/N	0/N	0/N	Endogenous peroxidase (granulocytes) and staining of mucus in all slides.	6	Positive	
	Hu87 ST-2	5	All elements	0/N	0/N	0/N	0/N	0/N	Endogenous peroxidase (granulocytes) present in all slides	6	Positive	
	Hu98 ST-2	5	All elements	0/N	0/N	0/N	0/N	0/N	Endogenous peroxidase (granulocytes) present in all slides	6	Positive	
Heart	Hu60 HT-4	5	All elements	0/N	0/N	0/N	0/N	0/N	Myocardium and epicardium present. Endogenous pigment (lipofuscin) present in all slides.	6	Positive	
	Hu74 HT-1	5	All elements	0/N	0/N	0/N	0/N	0/N	Myocardium and epicardium present. Endogenous pigment (lipofuscin) present in all slides.	6	Positive	
	Hu100 HT-2	5	All elements	0/N	0/N	0/N	0/N	0/N	Myocardium and epicardium present. Endogenous pigment (lipofuscin) present in all slides.	6	Positive	
Kidney (glomerulus, tubules)	Hu76 KD-2	5	All elements	0/N	0/N	0/N	0/N	0/N	No comments.	6	Positive	
	Hu111 KD-4	5	All elements	0/N	0/N	0/N	0/N	0/N	Staining of tubular epithelial cells (likely proximal straight tubules) in all slides attributed to binding by secondary or tertiary antibody.	6	Positive	
	Hu152 KD-1	5	All elements	0/N	0/N	0/N	0/N	0/N	Staining of tubular epithelial cells (likely proximal straight tubules) in all slides attributed to binding by secondary or tertiary antibody.	6	Positive	

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		0	Negative						N	Negative, no reactivity	
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		2	Mild						R	Rare, 5-25% of cells of that cell type	
		3	Moderate						O	Occasional, 26-50% of cells of that cell type	
		4	Strong						F	Frequent, 51-75% of cells of that cell type	
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		Relative Staining Strength by Location (tissue element or subcellular, when applicable):									
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Cryosection	Donor and Block	Run Number	Tissue Elements; Subcellular Localization	Specific Reactivity (Intensity/Frequency)					Tissue Comments	Vimentin	
				Test Article: LT3114-CF		Isotype Control Article: LT1023-CF		Assay Control		Run Number	Staining Results
				10 µg/mL	1 µg/mL	10 µg/mL	1 µg/mL				
Liver	Hu77 LI-2	5	All elements	0/N	0/N	0/N	0/N	0/N	Endogenous pigment (bilirubin) in all slides. Patchy staining of sinusoidal lining cells in all slides attributed to binding by secondary or tertiary antibody.	6	Positive
	Hu100 LI-5	5	All elements	0/N	0/N	0/N	0/N	0/N	Endogenous pigment (bilirubin) in all slides. Patchy staining of sinusoidal lining cells in all slides attributed to binding by secondary or tertiary antibody.	6	Positive
	Hu109 LI-6	5	All elements	0/N	0/N	0/N	0/N	0/N	Endogenous pigment (bilirubin) in all slides. Staining of biliary epithelium in all slides attributed to binding by secondary or tertiary antibody.	6	Positive
Lung	Hu28 LU-4	5	All elements	0/N	0/N	0/N	0/N	0/N	Exogenous pigment (carbon) in all slides. Endogenous pigment in alveolar macrophages in all slides. Variable staining of alveolar macrophages, and interstitial cells in all slides attributed to binding by secondary or tertiary antibody.	6	Positive
	Hu98 LU-7	5	All elements	0/N	0/N	0/N	0/N	0/N	Exogenous pigment (carbon) in all slides. Endogenous pigment in alveolar macrophages in all slides. Variable staining of alveolar macrophages, and interstitial cells in all slides attributed to binding by secondary or tertiary antibody.	6	Positive
	Hu100 LU-2	5	All elements	0/N	0/N	0/N	0/N	0/N	Exogenous pigment (carbon) in all slides. Endogenous pigment in alveolar macrophages in all slides. Variable staining of alveolar macrophages, and interstitial cells in all slides attributed to binding by secondary or tertiary antibody.	6	Positive
Lymph Node	Hu31 LN-5	5	All elements	0/N	0/N	0/N	0/N	0/N	No comments.	6	Positive
	Hu98 LN-3	5	All elements	0/N	0/N	0/N	0/N	0/N	Exogenous pigment (carbon), consistent with a bronchial lymph node.	6	Positive
	Hu99 LN-1	5	All elements	0/N	0/N	0/N	0/N	0/N	No comments.	6	Positive
Ovary	Hu98 OV-1	7	All elements	0/N	0/N	0/N	0/N	0/N	Follicles present. Several inclusion cysts present (normal incidental finding). Fallopian tube also present on slide. Ovary consistent with post-menopausal ovary.	8	Positive
	Hu108 OV-5	7	All elements	0/N	0/N	0/N	0/N	0/N	Endogenous pigment (lipofuscin) in all slides. Fallopian tube present. Ovary consistent with post-menopausal ovary.	8	Positive
	Hu116 OV-5	7	All elements	0/N	0/N	0/N	0/N	0/N	Stromal tissue present, but no follicles or corpus luteum, consistent with post-menopausal ovary.	8	Positive

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Cryosection	Donor and Block	Run Number	Tissue Elements; Subcellular Localization	Specific Reactivity (Intensity/Frequency)					Tissue Comments	Vimentin	
				Test Article: LT3114-CF		Isotype Control Article: LT1023-CF		Assay Control		Run Number	Staining Results
				10 µg/mL	1 µg/mL	10 µg/mL	1 µg/mL				
Pancreas	Hu35 PA-1	7	All elements	0/N	0/N	0/N	0/N	0/N	Staining of acinar cell cytoplasm in all slides attributed to binding by secondary or tertiary antibody.	8	Positive
	Hu100 PA-6	7	All elements	0/N	0/N	0/N	0/N	0/N	No comments.	8	Positive
	Hu103 PA-2	7	All elements	0/N	0/N	0/N	0/N	0/N	Staining of acinar cell cytoplasm in all slides attributed to binding by secondary or tertiary antibody.	8	Positive
Parathyroid	Hu41 TH-7	11	All elements	0/N	0/N	0/N	0/N	0/N	Endogenous peroxidase (follicular epithelium). Parathyroid present on slide with thyroid.	12	Positive
	Hu69 TH-8	*16, 18	U	U	U	U	U	U	Replacement block for Hu98 TH-12. Parathyroid missing	17	Positive
	Hu98 TH-12	11	U	U	U	U	U	U	Parathyroid missing. Replaced with Hu69 TH-8.	12	Positive
	Hu123 TH-5	11	U	U	U	U	U	U	Parathyroid missing. Replaced with Hu146 TH-2.	12	Positive
	Hu146 TH-2	*16, 18	All elements	0/N	0/N	0/N	0/N	0/N	Replacement block for Hu123 TH-5. Endogenous peroxidase (follicular epithelium). Parathyroid present on slide with thyroid.	17	Positive
Peripheral nerve	Hu98 MU-4	9	All elements	0/N	0/N	0/N	0/N	0/N	Peripheral nerve present in block containing skeletal muscle.	10	Positive
	Hu138 SG-4	9	All elements	0/N	0/N	0/N	0/N	0/N	Peripheral nerve present in fascia of block containing salivary gland.	10	Positive
	Hu104 MG-1	15	All elements	0/N	0/N	0/N	0/N	0/N	Peripheral nerve present.	2	Positive
Pituitary	Hu67 PI-1	7	All elements	0/N	0/N	0/N	0/N	0/N	No comments.	8	Positive
	Hu141 PI-2	7	All elements	0/N	0/N	0/N	0/N	0/N	No comments.	8	Positive
	Hu147 PI-1	7	All elements	0/N	0/N	0/N	0/N	0/N	No comments.	8	Positive
Placenta	Hu78 PL-1	7	All elements	0/N	0/N	0/N	0/N	0/N	Amnion, basal plate, and chorionic villi present. Variable staining of endothelial cells of chorionic villi in all slides attributed to secondary or tertiary antibody.	8	Positive
	Hu80 PL-4	7	All elements	0/N	0/N	0/N	0/N	0/N	Amnion, basal plate, and chorionic villi present. Variable staining of endothelial cells of chorionic villi in all slides attributed to secondary or tertiary antibody.	8	Positive
	Hu81 PL-1	7	All elements	0/N	0/N	0/N	0/N	0/N	Amnion, basal plate, and chorionic villi present. Variable staining of endothelial cells of chorionic villi in all slides attributed to secondary or tertiary antibody.	8	Positive

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				Test Article: LT3114-CF		Isotype Control Article: LT1023-CF		Assay Control		Run Number	Staining Results
				10 µg/mL	1 µg/mL	10 µg/mL	1 µg/mL				
Prostate	Hu37 PR-7	7	All elements	0/N	0/N	0/N	0/N	0/N	No comments.	8	Positive
	Hu99 PR-7	7	All elements	0/N	0/N	0/N	0/N	0/N	No comments.	8	Positive
	Hu100 PR-9	7	All elements	0/N	0/N	0/N	0/N	0/N	No comments.	8	Positive
Salivary gland	Hu121 SG-2	9	All elements	0/N	0/N	0/N	0/N	0/N	Patchy cytoplasmic staining of duct and acinar epithelial cells in all slides attributed to secondary or tertiary antibody.	10	Positive
	Hu138 SG-4	9	All elements	0/N	0/N	0/N	0/N	0/N	Patchy cytoplasmic staining of duct and acinar epithelial cells in all slides attributed to secondary or tertiary antibody. Peripheral nerve present in fascia.	10	Positive
	Hu144 SG-4	9	All elements	0/N	0/N	0/N	0/N	0/N	No comments.	10	Positive
Skin	Hu55 SK-8	9	All elements	0/N	0/N	0/N	0/N	0/N	Endogenous pigment (melanin) in all slides.	10	Positive
	Hu62 SK-4	9	All elements	0/N	0/N	0/N	0/N	0/N	Endogenous peroxidase (mast cells) and endogenous pigment (melanin) in all slides.	10	Positive
	Hu107 SK-1	9	All elements	0/N	0/N	0/N	0/N	0/N	Staining of dermal gland epithelium in all slides attributed to binding by secondary and tertiary antibodies.	10	Positive
Spinal cord	Hu109 SC-1	9	All elements	0/N	0/N	0/N	0/N	0/N	Variable staining of spindloid cells in all slides attributed to binding by secondary or tertiary antibody.	10	Positive
	Hu117 SC-5	9	All elements	0/N	0/N	0/N	0/N	0/N	Endogenous pigment (melanin) in all slides.	10	Positive
	Hu131 SC-2	9	All elements	0/N	0/N	0/N	0/N	0/N	Endogenous pigment (lipofuscin) in all slides.	10	Positive
Spleen	Hu52 SP-2	9	All elements	0/N	0/N	0/N	0/N	0/N	Endogenous peroxidase (granulocytes) staining in all slides. Staining of cells within red pulp in all slides attributed to binding by secondary or tertiary antibody.	10	Positive
	Hu60 SP-5	9	All elements	0/N	0/N	0/N	0/N	0/N	Endogenous peroxidase (granulocytes) staining in all slides. Staining of cells within red pulp in all slides attributed to binding by secondary or tertiary antibody.	10	Positive
	Hu98 SP-5	9	All elements	0/N	0/N	0/N	0/N	0/N	Endogenous peroxidase (granulocytes) staining in all slides. Staining of cells within red pulp in all slides attributed to binding by secondary or tertiary antibody.	10	Positive
Striated (skeletal) muscle	Hu41 MU-3	9	All elements	0/N	0/N	0/N	0/N	0/N	No comments.	10	Positive
	Hu98 MU-4	9	All elements	0/N	0/N	0/N	0/N	0/N	Peripheral nerve present.	10	Positive
	Hu99 MU-8	9	All elements	0/N	0/N	0/N	0/N	0/N	No comments.	10	Positive

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				10 µg/mL	1 µg/mL	10 µg/mL	1 µg/mL				
Testis	Hu36 TE-7	9	All elements	0/N	0/N	0/N	0/N	0/N	Endogenous pigment, lipofuscin (Leydig cells) present in all slides.	10	Positive
	Hu37 TE-7	9	All elements	0/N	0/N	0/N	0/N	0/N	Endogenous pigment, lipofuscin (Leydig cells) present in all slides.	10	Positive
	Hu99 TE-3	9	All elements	0/N	0/N	0/N	0/N	0/N	Endogenous pigment, lipofuscin (Leydig cells) present in all slides.	10	Positive
Thymus	Hu118 TM-3	11	All elements	0/N	0/N	0/N	0/N	0/N	Moderate physiologic atrophy. Endogenous peroxidase (granulocytes). Occasional minimal staining of spindloid cells within adipose tissue in all slides.	12	Positive
	Hu142 TM-1	*16, 18	U	U	U	U	U	U	Potential replacement block for Hu142 TM-3. However, due to marked physiologic atrophy (no lymphoid tissue apparent), block 3 for this donor was used.	17	Positive
	Hu142 TM-3	11	All elements	0/N	0/N	0/N	0/N	0/N	Marked physiologic atrophy (no lymphoid tissue apparent). Attempted to replace with alternate block. Endogenous peroxidase (granulocytes).	12	Positive
	Hu149 TM-1	11	All elements	0/N	0/N	0/N	0/N	0/N	Moderate physiologic atrophy. Endogenous peroxidase (granulocytes) and minor nonspecific staining of squamous epithelium of Hassall's corpuscles and scattered spindloid cells in adipose tissue in all slides.	12	Positive
Thyroid	Hu41 TH-7	11	All elements	0/N	0/N	0/N	0/N	0/N	Endogenous peroxidase (follicular epithelium). Parathyroid present.	12	Positive
	Hu98 TH-12	11	All elements	0/N	0/N	0/N	0/N	0/N	Endogenous peroxidase (follicular epithelium). No parathyroid present.	12	Positive
	Hu123 TH-5	11	All elements	0/N	0/N	0/N	0/N	0/N	Endogenous peroxidase (follicular epithelium). No parathyroid present. Staining of spindloid cells in fascial tissue around thyroid in all slides attributed to binding by secondary or tertiary antibody.	12	Positive
Tonsil	Hu133 TO-1	11	All elements	0/N	0/N	0/N	0/N	0/N	Endogenous peroxidase (granulocytes) staining, and minimal, cytoplasmic staining of squamous epithelium in epithelial crypts in all slides.	12	Positive
	Hu137 TO-6	11	All elements	0/N	0/N	0/N	0/N	0/N	Endogenous peroxidase (granulocytes) staining in all slides.	12	Positive
	Hu145 TO-4	11	All elements	0/N	0/N	0/N	0/N	0/N	Endogenous peroxidase (granulocytes) staining, and some mild staining of spindloid cells in all slides attributed to binding by secondary or tertiary antibody.	12	Positive

NA = Not applicable;
U=Unavailable.

DGG14-010	Scoring Definitions	Intensity of specific reactivity							Frequency of cells displaying specific reactivity				
		0	Negative							N	Negative, no reactivity		
		1	Weak (blush)							VR	Very rare, <5% of cells of that cell type		
		2	Mild							R	Rare, 5-25% of cells of that cell type		
		3	Moderate							O	Occasional, 26-50% of cells of that cell type		
		4	Strong							F	Frequent, 51-75% of cells of that cell type		
		5	Intense							VF	Very frequent, 76-100% of cells of that cell type		
	Relative Staining Strength by Location (tissue element or subcellular, when applicable):												
	* Due to inadequate staining of positive control material, tissues stained in Runs 1 and 16 could not be evaluated for test article reactivity and were restained in subsequent runs, with the exception of tissues which were unavailable or unsuitable for staining.												
Cryosection	Donor and Block	Run Number	Tissue Elements; Subcellular Localization	Specific Reactivity (Intensity/Frequency)					Tissue Comments	Vimentin			
				Test Article: LT3114-CF		Isotype Control Article: LT1023-CF		Assay Control		Run Number	Staining Results		
				10 µg/mL	1 µg/mL	10 µg/mL	1 µg/mL						
Ureter	Hu109 UR-6	11	All elements	0/N	0/N	0/N	0/N	0/N	Minimal, nonspecific, cytoplasmic staining of squamous epithelial cells in all slides.	12	Positive		
	Hu110 UR-1	11	All elements	0/N	0/N	0/N	0/N	0/N	Endogenous peroxidase (granulocytes)	12	Positive		
	Hu113 UR-1	11	All elements	0/N	0/N	0/N	0/N	0/N	Endogenous peroxidase (granulocytes)	12	Positive		
Urinary bladder	Hu88 UB-1	11	All elements	0/N	0/N	0/N	0/N	0/N	Endogenous peroxidase (granulocytes). Staining of cytoplasmic granules in bladder epithelium in all slides attributed to binding by secondary or tertiary antibody.	12	Positive		
	Hu94 UB-2	11	All elements	0/N	0/N	0/N	0/N	0/N	Endogenous peroxidase (granulocytes)	12	Positive		
	Hu100 UB-2	11	All elements	0/N	0/N	0/N	0/N	0/N	Endogenous peroxidase (granulocytes)	12	Positive		
Uterus - body (endometrium)	Hu82 UT-1	11	All elements	0/N	0/N	0/N	0/N	0/N	Scattered cytoplasmic staining of glandular epithelium attributed to binding by secondary or tertiary antibody.	12	Positive		
	Hu98 UT-3	11	U	U	U	U	U	U	Endometrium missing (myometrium only). Replaced with alternate block (Hu83 UT-2).	12	Positive		
	Hu83 UT-2	*16, 18	All elements	0/N	0/N	0/N	0/N	0/N	Replacement block for Hu98 UT-3.	17	Positive		
	Hu107 UT-5	11	All elements	0/N	0/N	0/N	0/N	0/N	No comments.	12	Positive		
Uterus - cervix	Hu50 CX-4	13	All elements	0/N	0/N	0/N	0/N	0/N	No comments.	14	Positive		
	Hu84 CX-3	13	All elements	0/N	0/N	0/N	0/N	0/N	No comments.	14	Positive		
	Hu85 CX-3	13	All elements	0/N	0/N	0/N	0/N	0/N	No comments.	14	Positive		

Appendix B. Peer Review Statement



Peer Review Statement

Client Identification Number:	<u>AZ-366</u>	
Study Number:	<u>DGG14-010</u>	
Study Pathologist and Affiliation:	<u>Joan R Wicks, DVM, PhD, DAVCP</u>	<u>Alizée Pathology, LLC</u>
Peer Review Scientist and Affiliation:	<u>Naiche L Adler, PhD</u>	<u>Alizée Pathology, LLC</u>

Purpose of the Review:

Internal, protocol directed peer review.

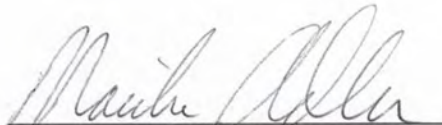
Scope of Peer Review:

Review of all tissue cross-reactivity (TCR) slides generated (staining runs 1 through 18) and pathology report narrative.

Peer Review Outcome:

- ☒ Peer review pathologist is in general agreement with the study pathologist regarding microscopic changes, terminology and interpretation.
- ☐ Peer review is incomplete and further evaluation is required.
- ☐ Peer review pathologist is in disagreement with the study pathologist on the following:

Comments: No comments.


Signature of Peer Review Scientist

1.5.2015
Date

Appendix C. Study Protocol and Amendment



Office: (240) 356-0056

Fax: (866) 929-5455

Alizée Pathology, LLC Final Tissue Cross-Reactivity Study Protocol

Alizée Study
Number DGG14-010

Study Title

Full Human Tissue Cross-Reactivity Study of LT3114

Testing
Facility Alizée Pathology, LLC
20 Frederick Road
Thurmont, MD 21788

Study Director Naiche L. Adler, PhD

Sponsor Lpath, Inc.
4025 Sorrento Valley Blvd.
San Diego, CA 92121

Date November 18, 2014

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I. Purpose

The purpose of this study is to evaluate potential tissue cross-reactivity of LT3114 on cryosections of normal human tissues.

II. Test Article Summary

The test article is a humanized IgG1 (κ) monoclonal antibody, LT3114, directed against lysophosphatidic acid (LPA).

III. Sponsor Information

A. Sponsor

Lpath, Inc.
4025 Sorrento Valley Blvd.
San Diego, CA 92121

B. Sponsor Representative

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GWoodnutt@lpath.com

C. Study Monitor

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mwmconsultinggroup@gmail.com

IV. Testing Facility

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20 Frederick Road
Thurmont, MD 21788

V. Testing Facility Personnel

A. Study Director

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B. Study Pathologist

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VI. Archive Facility and Archivist

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Document Center & Archive Manager
Calvert Laboratories, Inc.
Scott Technology Park
130 Discovery Drive
Scott Township, PA 18447
Telephone: 570-585-2215
terri.gyza@calvertlabs.com

VII. Protocol Changes

All changes in or revisions of this approved protocol and the reasons therefore shall be documented, signed by the study director, dated and maintained with the protocol. Deviations from the protocol will be described, explained, reviewed and dated in a timely fashion by the Study Director and maintained with study raw data.

VIII. Adherence to Regulatory Standards

A. Compliance

All portions of this study performed at the testing facility will adhere to the study protocol and Alizée Pathology's standard operating procedures (SOPs), and, except as noted below, the conduct of these portions of the study will conform to the most recent version of the following Good Laboratory Practice (GLP) standards: The Nonclinical Laboratory Studies Good Laboratory Practice Regulations issued by the U.S. Food and Drug Administration (Title 21 of the Code of Federal Regulations, Part 58).

Compliance of any portion of the study performed by the Sponsor, such as test and control article characterization including appropriate testing for identity, stability, strength, purity, composition or other characteristics for each batch or lot and methods of synthesis, fabrication, or derivation of the test and control articles and stability testing as applicable is the responsibility of the Sponsor.

Exceptions to the GLP compliant processes include:

- The optimization of test article concentrations, upon which the Validation Study, Alizée Study Number DGG14-010 (Pilot), was based, as well as other variable parameters of the immunohistochemistry methodology performed by the testing facility (Alizée Pathology) constitutes unaudited method development and was conducted in accordance with Alizée Pathology relevant SOPs, appropriate scientific methodology, and good scientific judgment.
- Analysis of test and control article concentration will not be conducted as specified under 21 CFR, Part 58.113 (Mixtures of Articles and Carriers). Initial concentration of test article and isotype control article is established by the Sponsor, as detailed above, and direct testing of final concentration(s) in carrier is impractical due to the admixture of other proteins and immunoglobulins. Mixture and dilution of the test article and isotype control article will be conducted according to relevant Alizée SOPs and study documentation and will be subject to quality assurance audit. Empirical validation of test article reactivity at final working dilutions will be verified with positive and negative control samples in each staining run. Retention samples of test article and isotype control article and all documentation necessary to reconstruct working dilutions will be retained for any necessary further analysis. Documentation of test and isotype control articles and dilutions used will also be subject to quality assurance audit.

In the opinion of the Study Director, these exceptions to GLP compliance will not impact the quality, integrity, or validity of this study. Additional exceptions to full GLP compliance, if any, will be documented in the raw data and in the compliance statement of the final report.

B. Quality Assurance

All data for this tissue cross-reactivity study and the final report will be audited by the Immunohistochemistry Testing Facility's Quality Assurance Unit (Alizée Pathology QAU) in accordance with the SOPs of Alizée Pathology. A quality assurance statement, including dates inspections were performed and reported to management, will be provided for inclusion in the final report. At least one of the following critical phases will be subject to inspection by Alizée Pathology QAU:

- Preparation and documentation of test and control article dilutions,
- Tissue sectioning,
- Immunohistochemistry procedure

IX. Proposed Study Timetable

- Study Initiation Date: Date the Study Director signs the appropriate signature line in section XVII, Signatures and Dates.
- Experimental Start Date: November 18, 2014
- Experimental End Date and Submission of Audited Draft Report: January 5, 2015

X. Test and Control Articles

A. Test Article

Description: LT3114 (test article) is a humanized monoclonal IgG1 (κ) antibody directed against lysophosphatidic acid (LPA) and has been conjugated to 5-(6)-carboxyfluorescein (CF), supplied at a concentration of 3.6 mg/mL, pH 7.4. The conjugated form of the test article is designated LT3114-CF.

Identification Number (Lot): GA121459.

Manufacturer: Test article is manufactured by Gallus-MO on behalf of Lpath Inc. CF conjugation was performed by Columbia Biosciences. The final concentration of the solution will be documented in the raw data.

Storage Conditions: Store at 2 to 8°C.

Stability: Stability of LT3114 is unknown. Information on stability under conditions of administration (room temperature) will be included in the final report if it has been determined.

Test Article Characteristics: Information on identity, purity, and composition, or other characteristics that define the Test Article are on file with the manufacturer. A Certificate of Analysis has been provided and the lot/batch of the test article will be retained in the working folder and included in the final report.

Handling Precautions: A Material Safety Data Sheet (MSDS) will be kept on file in the working folder for this study.

B. Isotype Control Article

Description: LT1023 (isotype control article), is a humanized non-specific monoclonal IgG1 (κ) antibody and has been conjugated to 5-(6)-carboxyfluorescein (CF), supplied at a concentration of 2.9 mg/mL, pH 7.4. The conjugated form of the isotype control article is designated LT1023-CF.

Identification Number: Lot number LP101210.

Manufacturer: Isotype control article is manufactured and supplied by Lpath, Inc. CF conjugation was performed by Columbia Biosciences. The final concentration of the solution will be documented in the raw data.

Storage Conditions: Store at 2 to 8°C.

Stability: Stability of LT1023 is unknown. Information on stability under conditions of administration (room temperature) will be included in the final report if it has been determined.

Control Article Characteristics: Information on identity, purity, stability, and composition, or other characteristics that define the control article are on file with the manufacturer. The lot/batch of the control article will be retained in the working folder and included in the final report.

Handling Precautions: An MSDS will be kept on file in the working folder for this study.

C. Safety Precautions

Standard laboratory safety procedures, in accordance with MSDSs, will be employed for handling the test and control articles and antibody diluents.

D. Retention Samples

Retention samples of the test and isotype control articles (one 50 µL aliquot each) will be taken by Alizée Pathology and stored under the same conditions as the aliquots used for experiments for possible analysis. Storage conditions are maintained and recorded in facility records, according to Alizée Pathology SOPs. If the storage conditions are not within the protocol specified ranges (from receipt until experiment) and the degree of deviation is deemed by the Study Director and Sponsor Representative to warrant investigation, the samples may be analyzed (to be added by amendment). If the retention samples are analyzed, replacement retention samples will be taken and stored as above, provided that sufficient quantities of test article and isotype control article remain. If the samples are not analyzed, they will be returned to Calvert Laboratories, Inc. for archive.

E. Shipping Information

The conjugated test article and isotype control article were shipped via client courier, temperature maintained, to the address indicated below:

Alizée Pathology
20 Frederick Road
Thurmont, MD 21788

F. Accountability and Disposition

Test article accountability will be maintained according to Alizée Pathology SOPs. Concurrent with finalization of the study, the remaining test and isotype control article will be reconciled and disposition will be based on direction from the Sponsor. The Sponsor has provided Alizée Pathology with statements of testing for the test article and isotype control article, and is responsible for providing details on stability and any other pertinent information derived from portions of the study conducted by the Sponsor.

XI. Test System (Cryosections of Human Tissues)

A. Source

Tissues were obtained at autopsy or biopsy from humans. Pertinent data (*i.e.*, identification, source, and age) for tissues used on study are on file at Alizée Pathology. Tissues as received are considered to be essentially normal based upon clinical history and review by Alizée pathologists. Each human donor is assigned a unique identification number that is documented in the working folder.

B. Justification for Selection of Test System

Tissue cross-reactivity studies are conducted as part of the investigational new drug (IND) submission for therapeutic antibodies and other biologics. The purpose of this study is to evaluate the potential reactivity of LT3114 on cryosections of normal human tissues. The test system has no requirements for bodyweight range, sex, or age of donors other than for selection of normal tissues and sex-specific tissues.

C. Test System Identification

Upon receipt at Alizée, donor tissue is identified with a unique permanent number which is recorded in the appropriate Master Tissue Bank and Individual Donor Inventory Sheet forms, on all relevant paperwork, and on the protective sealable and freezer-safe container. This number and any associated sub-numbers will be used as reference throughout the study.

D. Tissue Embedding and Storage

Unfixed tissue samples are placed in molds and frozen on dry ice in optimal cutting temperature compound (OCT) embedding medium. Tissues are placed in sealed plastic bags and maintained below -70°C until sectioning.

E. Tissue Sectioning

Tissue sectioning will be performed according to Alizée Pathology's Cryotomy SOP. Sections will be cut at approximately 4-5 μm , unless directed otherwise by the Study Pathologist, and post-fixed for 10 minutes in acetone followed by 10 seconds in 10% neutral buffered formalin (NBF).

F. Test Tissues

The selection of tissues to be tested is based on recommendations in Annex II of Directive 75/318/EEC, "Production and Quality Control of Monoclonal Antibodies" and in Appendix I of the 1997 US FDA/CBER "Points to Consider in the Manufacture and Testing of Monoclonal Antibody Products for Human Use."

Test tissues will consist of specimens of the following tissues from three different human donors, where available:

Adrenal	Liver	Skin
Blood cells*	Lung	Spinal cord
Blood vessels (endothelium)**	Lymph node	Spleen
Bone marrow	Ovary	Striated (skeletal) muscle
Brain – cerebrum (cortex), cerebellum	Fallopian tube (oviduct)	Testis
Breast (mammary gland)	Pancreas	Thymus
Eye	Parathyroid	Thyroid
Gastrointestinal tract – colon (large intestine)	Peripheral nerve	Tonsil
Gastrointestinal tract – esophagus	Pituitary	Ureter
Gastrointestinal tract – small intestine	Placenta	Urinary bladder
Gastrointestinal tract – stomach	Prostate	Uterus – body (endometrium)
Heart	Salivary (parotid) gland	Uterus – cervix
Kidney (glomerulus, tubule)		

*Blood cells include granulocytes, lymphocytes, monocytes and platelets and will be evaluated on a blood smear.

**Endothelium will be evaluated in all tissues.

NOTE: If a specific specimen cannot be acquired in a timely fashion, that tissue may not be evaluated in this study. Any occurrence of this sort will be noted in the final report, and the impact on the study will be addressed.

G. Method Controls

- Positive Control: LPA(C12)-SMCC-BSA spotted to slide. Spot slides are prepared by diluting 0.05 µg of LPA(C12)-SMCC-BSA (positive control material) in carbonate buffer (100 mM NaHCO₃, 33.6 Na₂CO₃, pH 9.5), applying 10 µL amounts to glass microscope slides, and drying the mixture on a glass slide at 37°C for 30 minutes.
- Negative Control: S1P-SMCC-BSA spotted to slide. Spot slides are prepared by diluting 0.05 µg of S1P-SMCC-BSA (negative control material) in carbonate buffer, applying 10 µL amounts to glass microscope slides, and drying the mixture on a glass slide at 37°C for 30 minutes.
- Isotype Control Article: LT1023.
- Assay Control: The assay control will act as the control for all aspects of the staining procedure aside from the primary antibody (test article or isotype control article). The immunoperoxidase staining method will be identical to that used for the test article except that dilution buffer will be used as a replacement for the primary antibody.
- Tissue Staining Control: The adequacy of all tissues used in this study (with the exception of the positive and negative control tissues and the blood smear) will be validated through immunohistochemical staining for the presence of vimentin. Vimentin is an intracellular structural protein expressed by many cell types and is present in all tissues used for this study.

XII. Immunoperoxidase Staining

A. Rationale for Method Selection

In the optimization and validation staining runs, the positive control material (LPA[C12]-SMCC-BSA) displayed staining with the CF-conjugated test article at concentrations between 0.1 and 30 µg/mL, with a drop-off in staining intensity and frequency at concentrations below 1 µg/mL. Therefore, the ideal concentration (*i.e.*, the lowest concentration of CF-conjugated test article giving no reduction in staining intensity or frequency) was determined to be 1 µg/mL. There was no staining of the positive control material with the CF-conjugated isotype control antibody and no staining of the negative control material with the CF-conjugated test or isotype control articles, nor with the assay control conditions. There was a minor increase in nonspecific staining at concentrations above 10 µg/mL. Therefore, the highest concentration of test article proposed for the human cross-reactivity study was determined to be a 10-fold increase over the ideal concentration, *i.e.*, 10 µg/mL. The staining concentrations for the CF-conjugated test article and isotype control article proposed for the GLP human tissue cross-reactivity study are 1 and 10 µg/mL.

B. Working Antibody Concentrations for Testing

The staining concentrations for the test article and isotype control article proposed for the GLP human tissue cross-reactivity study are 1 and 10 µg/mL.

C. Reagents

Reagents for Test Article and Isotype Control Article Staining

- Biocare Tris-buffered saline Auto-Wash (TBS-AW) (*Biocare Medical*)
- Peroxidized® 1 (*Biocare Medical*)
- Peroxidized® 1 Diluent (*Biocare Medical*)
- Human γ-globulins (HGG) (*Sigma-Aldrich, catalog number I4506, lot number 121M7009V*)
- Avidin/Biotin blocking kit (*Vector Laboratories*)
- Mouse monoclonal anti FITC (*Sigma Aldrich, catalog number F5636, lot number 111M4794*)
- 14974 antibody diluent (provided by Calvert Laboratories, Inc.)
- Dako EnVision+ System–HRP (DAB, anti-mouse) (*Dako, catalog number K4007, lot number 10088650*)
- Sigma Fast 3,3' Diaminobenzidine (DAB) Solution (*Sigma-Aldrich, catalog number D4293, lot number SLBJ3609V*)
- Neutral Buffered Formalin (NBF)
- Acetone (*Mercedes Medical*)

Reagents for Staining with Anti-Vimentin Antibody

- Avidin/Biotin Blocking Kit (*Vector Laboratories*)
- Peroxidized® 1 (*Biocare Medical*)
- Peroxidized® 1 Diluent (*Biocare Medical*)
- Sniper® (*Biocare Medical*)
- Anti-human vimentin antibody (*Biocare Medical*)
- Goat anti-Mouse IgG – Biotinylated (*Jackson*)
- ABC Elite (*Vector Laboratories*)
- Sigma Fast DAB (*Sigma-Aldrich*)
- DaVinci Green Diluent (*Biocare Medical*)
- Biocare TBS Auto-wash (*Biocare Medical*)
- 10% Neutral Buffered Formalin (NBF)
- Acetone (*Mercedes Medical*)

D. Antibody Diluents

Primary and secondary diluent for the test article and isotype control article antibodies reaction sequence will be 14974 (24.3 mM citrate, 51.4 mM sodium phosphate, 300 mM D-trehalose pH 5.0 containing 2 mL 10% polysorbate 80), provided to Alizée by Calvert Laboratories, Inc. The anticipated volume of diluent needed for the length of the study has been provided in bulk. Diluent will be stored at 4°C. Use of the primary and secondary antibody diluent will be documented in the working folder (using Alizée Form 825-2, Frozen Immunohistochemistry Run Sheet).

E. Reaction Sequence: Test and Isotype Control Antibodies

An indirect immunoperoxidase procedure will be performed.

Antibody dilution:

1. Primary antibody and diluent at 1:3.33 ratio:
 - a. LT3114-CF at 1 or 10 µg/mL , or
 - b. LT1023-CF at 1 or 10 µg/mL, or
 - c. Primary diluent only for assay control.
2. Secondary antibody, mouse monoclonal anti-FITC, and diluent at 1:500 ratio.

Staining Procedure

1. Fix slides in cold acetone for 10 minutes and allow to air dry.
2. Fix slides for 10 seconds in 10% NBF.
3. Rinse twice with TBS-AW.
4. Apply Avidin complex for 15 minutes.
5. Rinse once with TBS-AW.
6. Apply Biotin complex for 15 minutes.
7. Rinse once with TBS-AW.
8. Apply 1 mg/mL heat aggregated human IgG in PBS, 0.5% Casein and 5% Human Gamma Globulins for 30 minutes.
9. Add primary antibody.
10. Incubate for 60 minutes at room temperature.
11. Wash twice with TBS-AW
12. Apply 1:5 ratio of Peroxidized 1 with Peroxidized 1 Diluent.
13. Rinse twice with TBS-AW.
14. Add Mouse monoclonal anti-FITC for 30 minutes.
15. Wash twice with TBS-AW.
16. Apply Dako envision mouse HRP polymer for 30 minutes.
17. Wash twice with TBS-AW.
18. Apply Sigma Fast DAB solution for 5 minutes for color development.
19. Rinse with distilled water.
20. Counterstain with hematoxylin and coverslip the slides according to Alizée SOPs

F. Validation of Tissue Samples (Anti-Vimentin Staining): Reaction Sequence

Viability of tissue samples for immunohistochemical staining will be validated by anti-vimentin staining using the following reaction sequence:

1. Fix slides in cold acetone for 10 minutes and allow to air dry.
2. Fix slides for 10 seconds in 10% neutral buffered formalin (NBF).
3. Place in distilled water for 2 minutes.
4. Rinse twice with Biocare TBS-AW.
5. Apply Avidin block for 15 minutes.
6. Rinse once with TBS-AW.
7. Apply Biotin block for 15 minutes.
8. Rinse once with TBS-AW.
9. Apply 1:5 Peroxidized I with Peroxidized 1 Diluent for 5 minutes to reduce endogenous peroxidase staining.
10. Rinse twice with TBS-AW.
11. Apply protein block, Sniper, for 20 minutes to reduce nonspecific binding.
12. Rinse twice with TBS-AW.
13. Apply primary antibody, 1:500 anti-vimentin to DaVinci Green, and incubate for 1 hour at room temperature.
14. Rinse twice with TBS-AW.
15. Apply secondary antibody, 1:500 biotinylated anti-mouse IgG, for 30 minutes.
16. Rinse twice with TBS-AW.
17. Apply ABC Elite reagent for 30 minutes.
18. Rinse twice with TBS-AW.
19. Apply Sigma Fast DAB solution and for color development and incubate for 5 minutes.
20. Rinse in distilled water
21. Counterstain and coverslip the slides according to Alizée SOPs.

G. Verification

If deemed appropriate by the Study Director, verification staining runs may be conducted during the course of the study. The tissues stained in the verification staining run will include positive and negative control tissues and may include a small number of test tissues. Reasons for a verification staining run will be documented in the working folder as a memorandum to the Working Folder. Details of the exact method and staining dilutions (concentrations) to be used will be included in the Final Study Report.

H. Test and Measurements

The Study Pathologist will evaluate all immunohistochemically stained slides, including negative control slides and assay control slides and will determine the adequacy of tissue morphology and staining. Data reporting will include identification of the tissue or cell type showing reactivity with the test article, and frequency and average intensity of that staining. If appropriate, the subcellular localization of staining will also be reported. The intensity of staining will be graded as follows: 0 = negative; 1 = weak (blush); 2 = mild; 3 = moderate; 4 = strong; 5 = intense.

The frequency of staining will be graded according to the approximate percentage of a particular cell type or tissue element as follows: N = negative; VR = very rare, less than 5%; R = rare, 5-25%; O = occasional, 26-50%; F = frequent, 51-75%; VF = very frequent, 76-100%. All data will be recorded in Microsoft® Excel. The Alizée Pathology Peer Review Scientist will examine all positive and negative control slides, all slides stained with test article, and selected slides stained with isotype control article or assay control, and will review the report. Peer review will be documented on Alizée's Peer Review Statement.

As the staining produced by test article solutions will be evaluated in the context of isotype control article and assay control staining solutions, control of bias by blinding or other methods is not appropriate for this study. Due to the semiquantitative nature of the data, statistical analysis will not be performed.

XIII. Records

The following will be collected and retained according to Alizée SOPs:

- Test article and isotype control article records.
- Retention samples of the test article and isotype control articles.
- Study slides.
- Letters or any written communication (including emails) concerning the study that are required for reconstruction of the study.
- Protocol, amendments and deviations.
- Documentation concerning study procedures.
- Quality assurance inspection reports.
- Final report.

XIV. Report

A complete detailed audited draft report will be generated. Only the signed hard copy of the final report will be considered raw data. The report will include, but will not necessarily be limited to, materials and experimental methods, results of all portions of the study conducted at Alizée Pathology, including data tables, and summary. The study protocol, protocol amendments, a quality assurance statement, a list of any deviations from regulations cited in section VIII., Adherence to Regulatory Standards, and documentation of the identity, strength, and composition of the test article and isotype control article will be included as appendices to the report.

XV. Material Disposition

Concurrent with completion of the study, all items listed in XIII., Records, will be archived in compliance with all applicable regulations at Calvert Laboratories, Inc. (see VI., Archive Facility and Archivist).

XVI. References

1. Annex II Directive 75/318/EEC, "Production and Quality Control of Monoclonal Antibodies," last revised December 1994.
2. "Points to Consider in the Manufacture and Testing of Monoclonal Antibody Products for Human Use," U.S. Department of Health and Human Services, Food and Drug Administration, Center for Biologics Evaluation and Research. February 28, 1997.

XVII. Signatures and Dates

A. Sponsor Representative

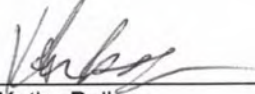
Gary Woodnutt
Sponsor Representative

Date

B. Testing Facility Responsible Personnel

Naiche L. Adler
Study Director

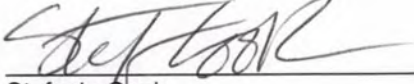
Date



Kathy Bailey
Testing Facility Management

18 Nov 14

Date



Stefanie Cook
Testing Facility Quality Assurance

18 Nov 14

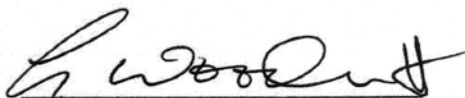
Date

XVI. References

1. Annex II Directive 75/318/EEC, "Production and Quality Control of Monoclonal Antibodies," last revised December 1994.
2. "Points to Consider in the Manufacture and Testing of Monoclonal Antibody Products for Human Use," U.S. Department of Health and Human Services, Food and Drug Administration, Center for Biologics Evaluation and Research. February 28, 1997.

XVII. Signatures and Dates

A. Sponsor Representative



Gary Woodnutt
Sponsor Representative

18th November 2014
Date

B. Testing Facility Responsible Personnel



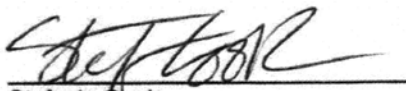
Naiche L. Adler
Study Director

18 Nov 14
Date



Kathy Bailey
Testing Facility Management

18 Nov 14
Date



Stefanie Cook
Testing Facility Quality Assurance

18 Nov 14
Date

Appendix A. DGG14-010 (Pilot) Executive Summary Immunohistochemistry Report



Office: (240) 356-0056

Fax: (866) 929-5455

Alizée Pathology, LLC

Executive Summary Immunohistochemistry Report

Alizée Study Number

DGG14-010 (Pilot)

Study Title

Method Development and Validation for
Immunohistochemistry with
Humanized Monoclonal Antibody
LT3114

Prepared By

Alizée Pathology, LLC
20 Frederick Road
Thurmont, MD 21788

Prepared For

Calvert Laboratories, Inc.
Scott Technology Park
130 Discovery Drive
Scott Township, PA 18447

Sponsor

Lpath, Inc.
4025 Sorrento Valley Blvd
San Diego, CA 92121

Study Director

Joan R. Wicks, DVM, PhD, DACVP

Date

11/14/2014

Alizée
Pathology, LLC

Executive Summary Immunohistochemistry Report
Method Development and Validation for Immunohistochemistry with
Humanized Monoclonal Antibody LT3114
Lpath, Inc.
Study Number DGG14-010 (Pilot)

This immunohistochemistry method development and validation executive summary by Alizée Pathology, LLC (Alizée) presents a summary of the immunohistochemistry method development for LT3114 for Calvert Laboratories, Inc., Scott Technology Park, 130 Discovery Drive, Scott Township, PA 18447.

The purpose of this pilot study was to optimize an immunohistochemical staining procedure to be used on human tissues for testing of tissue cross-reactivity with LT3114. The test article, LT3114, a humanized monoclonal IgG1 (κ) antibody directed against lysophosphotidic acid [LPA], along with an isotype control antibody, LT1023, were provided to Alizée, 20 Frederick Road, Thurmont, MD 21788, by the Sponsor, Lpath, Inc., via Calvert Laboratories, Inc. Conjugation of the test and isotype control articles to 5-(6)-carboxyfluorescein (CF) was performed at Columbia Biosciences, Frederick, MD.

Methods

The portions of Alizée Project Number DGG14-010 (Pilot) performed at Alizée were neither intended nor required to be in compliance with United States Food and Drug Administration Good Laboratory Practices (GLP) regulations set forth in Title 21 Code of Federal Regulations (CFR) Part 58. As such, quality assurance inspections and oversight were not performed. However, all study conduct performed at Alizée was completed in accordance with Alizée's Standard Operating Procedures (SOPs) and sound scientific judgment. The method development and validation results and the validated staining procedure presented in this report have undergone a thorough quality control review and accurately reflect the raw data.

The test article, LT3114 (*Lot number GA121459*), and isotype control article, LT1023 (*Lot number LP101210*) were provided by the Sponsor via Calvert Laboratories, Inc. A series of staining runs were performed utilizing a range of conditions and antibody concentrations in order to determine the optimal antibody concentrations for maximal positive control staining and minimal background staining. Method development staining runs were performed using an indirect immunoperoxidase procedure. Due to the potential reaction of the anti-human IgG (HulGg) secondary antibody with endogenous HulGg in the human study tissue, approaches to method development were initially performed by precomplexing the primary and secondary biotinylated antibodies overnight, followed by application of the precomplexed mixture to the slides. However, due to excessive nonspecific staining using this approach, an alternate approach was employed, involving conjugation of 5,6-carboxyfluorescein (CF), followed by a mouse anti-fluorescein secondary antibody, and a horseradish peroxidase polymer conjugated anti-mouse tertiary antibody (Dako EnVision+ System–HRP). Detection of binding was performed by incubating the slides with DAB, followed by counterstaining with hematoxylin. Conjugation of the test and isotype control antibodies to CF was performed by Columbia Biosciences.

LPA(C12)-SMCC-BSA adhered to microscope slides served as the positive control. Positive and negative control slides were prepared by diluting 0.05 µg of LPA(C12)-SMCC-BSA (positive control material) or S1P-SMCC-BSA (negative control material) in carbonate buffer (100 mM NaHCO₃, 33.6 Na₂CO₃, pH 9.5), applying 10 µL amounts to glass microscope slides, and drying the slides at 37°C for 30 minutes. Control slides (positive and negative) were subject to identical staining procedures as that proposed for study tissues, i.e. post-fixation in cold acetone for 10 minutes followed 10 seconds in 10% neutral buffered formalin at room temperature.

Method development and validation runs were performed at concentrations of CF-conjugated test article and isotype control article ranging from 0.1 to 30 µg/mL. Dilution buffer for the test and isotype control articles was used in place of test article or isotype control article as an assay control. All slides were evaluated via light microscopy by Joan R. Wicks, DVM, PhD, DACVP.

Results

In the methods development and validation staining runs, using an indirect immunoperoxidase technique with CF conjugated test and isotype control antibodies, an anti-fluorescein mouse IgG secondary, and a Dako Envision+ tertiary system, the positive control material (LPA[C12]-SMCC-BSA adhered to microscope slides) displayed staining with the CF-conjugated test article at concentrations between 0.1 and 30 µg/mL, with a drop-off in staining intensity and frequency at concentrations below 1 µg/mL. Therefore, the ideal concentration determined for the study (i.e. the lowest concentration of CF-conjugated test article giving no reduction in staining intensity or frequency) was 1 µg/mL. There was no staining of the positive control material with the CF-conjugated isotype control antibody and no staining of the negative control material, S1P-SMCC-BSA, with the CF-conjugated test or isotype control articles, or with the assay control. Due to the presence of minor nonspecific staining in tissue samples at 10 µg/mL, the higher concentration of CF-conjugated test article proposed for the human cross-reactivity study will be a 10-fold increase over the ideal concentration, i.e. 10 µg/mL. Thus, the staining concentrations for the CF-conjugated test and isotype control articles proposed for the GLP human tissue cross-reactivity study are 1 and 10 µg/mL.

Validated Staining Method

Based upon validation of the CF conjugated test and isotype control articles as described above, the following staining procedure will be used for the full tissue human GLP tissue cross-reactivity study DGG14-010.

An indirect immunoperoxidase procedure will be performed. 14974 diluent (24.3 mM citrate, 51.4 nM sodium phosphate, 300 mM D-trehalose pH 5.0 containing 2 mL 10% polysorbate 80), provided to Alizée by Calvert Laboratories, Inc., will serve as the diluent for primary and secondary antibodies. Based upon optimization and validation of staining conditions, the following reagents and procedure have been determined to be optimal for subsequent tissue cross-reactivity testing of the CF conjugated test article, LT3114-CF:

Validated Reagents:

- Biocare Tris-buffered saline Auto-Wash (TBS-AW) (*Biocare Medical*)
- Peroxidazed® 1 (*Biocare Medical*)
- Peroxidazed® 1 Diluent (*Biocare Medical*)
- Human γ-globulins (HGG) (*Sigma-Aldrich, catalog number I4506, lot number 121M7009V*)
- Avidin/Biotin blocking kit (*Vector Laboratories*)
- Mouse monoclonal anti FITC (*Sigma Aldrich, catalog number F5636, lot number 111M4794*)
- 14974 antibody diluent (provided by Calvert Laboratories, Inc.)
- Dako EnVision+ System–HRP (DAB, anti-mouse) (*Dako, catalog number K4007, lot number 10088650*)
- Sigma Fast 3,3' Diaminobenzidine (DAB) Solution (*Sigma-Aldrich, catalog number D4293, lot number SLBJ3609V*)
- Neutral Buffered Formalin (NBF)
- Acetone (*Mercedes Medical*)

Validated Staining Procedure:

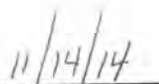
Staining Procedure

1. Fix slides in cold acetone for 10 minutes and allow to air dry.
2. Fix slides for 10 seconds in 10% NBF.
3. Rinse twice with TBS-AW.
4. Apply Avidin complex for 15 minutes.
5. Rinse once with TBS-AW.
6. Apply Biotin complex for 15 minutes.
7. Rinse once with TBS-AW.
8. Apply 1 mg/mL heat aggregated human IgG in PBS, 0.5% Casein and 5% Human Gamma Globulins for 30 minutes.
9. Add primary antibody.
10. Incubate for 60 minutes at room temperature.
11. Wash twice with TBS-AW
12. Apply 1:5 ratio of Peroxidized 1 with Peroxidized 1 Diluent.
13. Rinse twice with TBS-AW.
14. Add Mouse monoclonal anti-FITC for 30 minutes.
15. Wash twice with TBS-AW.
16. Apply Dako envision mouse HRP polymer for 30 minutes.
17. Wash twice with TBS-AW.
18. Apply Sigma Fast DAB solution for 5 minutes for color development.
19. Rinse with distilled water.
20. Counterstain with hematoxylin and coverslip the slides according to Alizée SOPs

Conclusion

In the optimization and validation staining runs, the positive control material (LPA[C12]-SMCC-BSA) displayed staining with the CF-conjugated test article at concentrations between 0.1 and 10 µg/mL, with a drop-off in staining intensity and frequency at concentrations below 1 µg/mL. Therefore, the ideal concentration (*i.e.*, the lowest concentration of CF-conjugated test article giving no reduction in staining intensity or frequency) was determined to be 1 µg/mL. There was no staining of the positive control material with the CF-conjugated isotype control antibody and no staining of the negative control material with the CF-conjugated test or isotype control articles, nor with the assay control conditions. There was a minor increase in nonspecific staining at concentrations above 10 µg/mL. Therefore, the highest concentration of test article proposed for the human cross-reactivity study was determined to be a 10-fold increase over the ideal concentration, *i.e.*, 10 µg/mL. The staining concentrations for the test article and isotype control article proposed for the GLP human tissue cross-reactivity study are 1 and 10 µg/mL.


Joan R. Wicks, DVM, PhD, DACVP
Study Pathologist


Date



STUDY TITLE: Full Human Tissue Cross-Reactivity Study of LT3114

STUDY NUMBER: DGG14-010

AMENDMENT NUMBER: 1

1. Subject: Page 3, I. Purpose

The purpose of this study is to evaluate potential tissue cross-reactivity of ~~LT3114~~ **LT3114-CF** on cryosections of normal human tissues.

Justification: Clarification of test article identity.

2. Subject: Page 3, II. Test Article Summary

The test article is a humanized IgG1 (κ) monoclonal antibody, ~~LT3114~~ **LT3114-CF**, directed against lysophosphatidic acid (LPA) **and has been conjugated to 5-(6)-carboxyfluorescein (CF)**.

Justification: Clarification of test article identity and description.

3. Subject: Page 6, X. Test and Control Articles, A. Test Article

Description: ~~LT3114~~ **LT3114-CF** (test article) is a humanized monoclonal IgG1 (κ) antibody directed against lysophosphatidic acid (LPA) and has been conjugated to 5-(6)-carboxyfluorescein (CF), supplied at a concentration of 3.6 mg/mL, pH 7.4. ~~The conjugated form of the test article is designated LT3114-CF.~~ Identification Number (Lot): ~~GA121459~~ **FAM999-14-005**.

Manufacturer: ~~The unconjugated test article antibody (LT3114)~~ **The unconjugated test article antibody (LT3114)** is manufactured by Gallus-MO on behalf of Lpath Inc. CF conjugation was performed by Columbia Biosciences. The final concentration of the solution will be documented in the raw data.

Storage Conditions: Store at 2 to 8°C.

Stability: Stability of **LT3114-CF and** LT3114 is unknown. Information on stability under conditions of administration (room temperature) will be included in the final report if it has been determined.

Test Article Characteristics: Information on identity, purity, and composition, or other characteristics that define the Test Article are on file with the manufacturer. A Certificate of Analysis has been provided and the lot/batch of the test article will be retained in the working folder and included in the final report.

Handling Precautions: A Material Safety Data Sheet (MSDS) will be kept on file in the working folder for this study.

Justification: Clarification of test article identity and description.

4. Subject: Page 7, X. Test and Control Articles, B. Isotype Control Article

Description: ~~LT1023~~ **LT1023-CF** (isotype control article), is a humanized non-specific monoclonal IgG1 (κ) antibody and has been conjugated to 5-(6)-carboxyfluorescein (CF), supplied at a concentration of 2.9 mg/mL, pH 7.4. ~~The conjugated form of the isotype control article is designated LT1023-CF.~~

Identification Number: Lot number ~~LP101210~~ **FAM999-14-006**.

Manufacturer: ~~The unconjugated isotype control article antibody (LT1023)~~ **The unconjugated isotype control article antibody (LT1023)** is manufactured and supplied by Lpath, Inc. CF conjugation was performed by Columbia Biosciences. The final concentration of the solution will be documented in the raw data.

Storage Conditions: Store at 2 to 8°C.

Stability: Stability of **LT1023-CF and LT1023** is unknown. Information on stability under conditions of administration (room temperature) will be included in the final report if it has been determined.

Control Article Characteristics: Information on identity, purity, stability, and composition, or other characteristics that define the control article are on file with the manufacturer. The lot/batch of the control article will be retained in the working folder and included in the final report.

Handling Precautions: An MSDS will be kept on file in the working folder for this study.

Justification: Clarification of isotype control article identity and description.

5. Subject: Page 8, XI. Test System (Cryosections of Human Tissues), B. Justification for Selection of Test system

Tissue cross-reactivity studies are conducted as part of the investigational new drug (IND) submission for therapeutic antibodies and other biologics. The purpose of this study is to evaluate the potential reactivity of ~~LT3114~~ **LT3114-CF** on cryosections of normal human tissues. The test system has no requirements for bodyweight range, sex, or age of donors other than for selection of normal tissues and sex-specific tissues.

Justification: Clarification of test article identity.

6. Subject: Page 9, XI. Test System (Cryosections of Human Tissues), G. Method Controls

- Isotype Control Article: ~~LT1023~~ **LT1023-CF**.

Justification: Clarification of isotype control article identity.

7. Subject: Page 10, Section XII. Immunoperoxidase Staining, C. Reagents

Reagents for Test Article and Isotype Control Article Staining

- Biocare Tris-buffered saline Auto-Wash (TBS-AW) (*Biocare Medical*)
- Peroxidazed® 1 (*Biocare Medical*)
- Peroxidazed® 1 Diluent (*Biocare Medical*)
- ~~Human γ -globulins (HGG)~~ **Human IgG (HulgG)** (*Sigma-Aldrich, catalog number I4506, lot number 121M7009V*)
- **Human γ -globulins (HGG) (Sigma-Aldrich, catalog number G4386, lot number SLBD8893V)**
- Avidin/Biotin blocking kit (*Vector Laboratories*)
- Mouse monoclonal anti FITC (*Sigma Aldrich, catalog number F5636, lot number 111M4794*)
- 14974 antibody diluent (provided by Calvert Laboratories, Inc.)
- Dako EnVision+ System–HRP (~~DAB~~ **peroxidase polymer**, anti-mouse) (*Dako, catalog number K4007, lot number 10088650*)
- Sigma Fast 3,3' Diaminobenzidine (DAB) Solution (*Sigma-Aldrich, catalog number D4293, lot number SLBJ3609V [5 mL] or catalog number D4414, lot number SLBJ3609V [15 mL]*)
- Neutral Buffered Formalin (NBF)
- Acetone (*Mercedes Medical*)

Justification: Correction of typographical errors, addition of reagent human γ -globulin, and addition of catalog and lot number for alternate volume of DAB reagent so that either can be used depending on required volume. Information verified by product labels and method development report.

8. Subject: Page 11, Section XII. Immunoperoxidase Staining, E. Reaction Sequence: Test and Isotype Control Antibodies

An indirect immunoperoxidase procedure will be performed.

Antibody dilution:

1. Primary antibody ~~solutions and diluent at 1:3.33 ratio:~~
 - a. LT3114-CF at 1 or 10 µg/mL , or
 - b. LT1023-CF at 1 or 10 µg/mL, or
 - c. Primary diluent only for assay control.
2. Secondary antibody, mouse monoclonal anti-FITC, and diluent at 1:500 ratio.

Staining Procedure

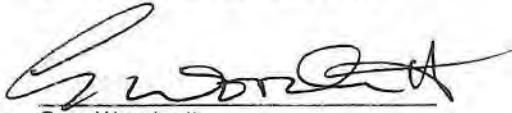
1. Fix slides in cold acetone for 10 minutes and allow to air dry.
2. Fix slides for 10 seconds in 10% NBF. **Place in distilled water for 2 minutes.**
3. Rinse twice with TBS-AW.
4. Apply Avidin complex for 15 minutes.
5. Rinse once with TBS-AW.
6. Apply Biotin complex for 15 minutes.
7. Rinse once with TBS-AW.
8. Apply 1 mg/mL heat aggregated human IgG in PBS, 0.5% Casein and 5% Human Gamma Globulins for 30 minutes.
9. Add primary antibody.
10. Incubate for 60 minutes at room temperature.
11. Wash twice with TBS-AW
12. Apply 1:5 ratio of Peroxidized 1 with Peroxidized 1 Diluent.
13. Rinse twice with TBS-AW.
14. Add Mouse monoclonal anti-FITC for 30 minutes.
15. Wash twice with TBS-AW.
16. Apply Dako envision mouse HRP polymer for 30 minutes.
17. Wash twice with TBS-AW.
18. Apply Sigma Fast DAB solution for 5 minutes for color development.
19. Rinse with distilled water.
20. Counterstain with hematoxylin and coverslip the slides according to Alizée SOPs

Justification: Removal of incorrect information and addition of procedural information in Step 2, which was inadvertently omitted from the original staining procedure. Changes verified against correct method development report.

Signatures and Dates

Amendment effective: Date of Study Director signature below.

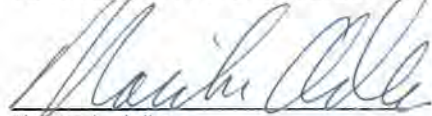
SPONSOR REPRESENTATIVE



Gary Woodnutt
Sponsor Representative

25th November 2014
Date

TEST FACILITY RESPONSIBLE PERSONNEL



Naiche L. Adler
Study Director

11.26.14
Date



Kathy Bailey
Testing Facility Management

25 Nov 2014
Date



Stefanie Cook
Testing Facility Quality Assurance

25 Nov 14
Date

Appendix D. Certificates of Analysis

290CT14

Lpath Inc.		Certificate of Analysis	
TITLE	Lpathomab-DS, 30 mg/mL, for use in GLP-conforming testing		REVISION No.
			1.0.0
			PAGE 1 of 2

Name		Part Number		Lot Number
Lpathomab-DS, 30 mg/mL		TBD		GA121459
Formulation Buffer				Storage Temperature
24.3 mM Sodium Citrate, 51.4 mM Sodium Phosphate, 300 mM D-Trehalose, 200 ppm Polysorbate-80, pH 5.0				2 to 8°C
Production Date		Test Date	Retest Date	Intended Use
JUN 2014		JUL 2014	DEC 2014	GLP-conforming Toxicity Testing
Manufacturer		Manufacturer Part Number		Manufacturer Lot Number
Gallus-MO, St. Louis, MO		TBD		Gallus-MO 103948
Attribute	Test	Specification		Results
Identity	Appearance	Particulate free, clear, colorless to pale yellow, slightly opalescent solution		Particulate free, clear, slightly yellow solution
	pH	5.0 ± 0.2		5.1
	CGE (Non-reduced)	Major band at 150 kDa		Conforms
	CGE (Reduced)	Two pronounced bands; One at ~50 kDa; One at ~ 25 kDa		Conforms
	iCE	Comparable to RS		Conforms
		Minor Peak at pI ~8.3		15%
Major Peak at pI ~8.4		60%		
Potency	API Concentration by OD at 280 nm	30 ± 3 mg/mL		32.1 mg/mL
	LPA Binding	100% ±40% of RS		89%

Lpath Inc.

Certificate of Analysis

TITLE

Lpathomab-DS, 30 mg/mL, for use in GLP-conforming testing

REVISION No.

1.0.0

PAGE

2 of 2

Attribute	Test	Specifications		Results
Purity	CGE non-reduced	Monomer + Non-glycosylated Monomer	≥ 90%	98.8%
		Non-glycosylated Monomer	Report %	2.6%
		Other impurities	No single impurity > 5%	conforms
	CGE, reduced	HC + LC + Non-Glycosylated HC	≥ 90%	99.0%
		HC	Report %	66.6%
		Non-Glycosylated HC	Report %	1.0%
		LC	Report %	31.4%
		Other Impurities	No single impurity > 5%	0.4% (high molecular weight)
	SE-HPLC	Purity [%]	≥ 90%	99.5%
		Aggregates [%]	≤ 5%	0.5%
		Other Impurities [%]	No single impurity > 5%	None detected
Attribute	Test	Specification		Results
Safety	Residual Protein-A	≤ 10 ppm		<1 ppm
	Residual DNA	≤ 5 pg/mg		<0.05 pg/mg
	CHO HCP	≤ 100 ppm		15 ppm
	Endotoxin, USP <85>	Report		<0.06 EU/mL ¹
	Bioburden; USP <61> (Membrane Filtration)	≤ 5 CFU/10 mL		None detected

JUDGMENT

Based on a comprehensive review of the lot specific manufacturing and testing records, this API is approved for use in GLP-conforming pre-clinical animal toxicity tests.

Caution! Not for use in humans!

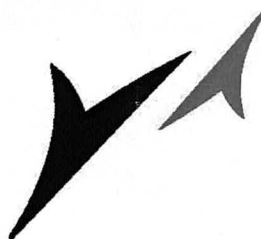
Quality Assurance:


 Acting Head of QA - Frieder K. Hofmann, PhD

Date: 22AUG2014

¹ At 0.06 EU/mL and a tolerance of 5 EU/kg animal weight, the maximum volume that may be dosed is 83 mL/kg or 2,675 mg Lpathomab per kg of animal.

Confidential



H14176 Received 10/14/14 RH

H14133 conjugated to 5-(6)-Carboxyfluorescein

Product Number	FAM-999
Lot Number	FAM999-14-005
Protein	Lpathomab DS, Lot 103948
5(6)-FAM SE	Biotium 90024, Lot 7F0730
Volume	2.35 ml
Conc. (mg)	3.60 mg/mL
Amount	8.5 mg
F:P	1.9
Store at	2-8°C

Form/Shipping & Storage

Supplied as liquid. Upon receipt, store at 2-8°C.

Handling

We recommend that the investigator determine the appropriate working concentration for their specific application. Avoid exposure to heat and light.

Buffer

The product is provided in PBS, pH 7.4



COLUMBIA BIOSCIENCES

4985 Winchester Blvd. Frederick, MD 21703 p: 310.732.5415 f: 443.558.3883 www.columbiabiosciences.com

Certificate of Analysis

TITLE	LT1023, human NS IgG antibody (15.3 mg/ml)				LOT NUMBER
					LP101210
DOCUMENT NUMBER	REVISION	SUPERSEDED REVISION	ISSUE DATE	EFFECTIVE DATE	PAGE
AB-COA-007	0.0.0	0.0.0	MAY 23, 2008	MAY 23, 2008	1 of 3

Name	Identity	Lot Number
LT1023	Human, non-specific, monoclonal IgG1k antibody	LP101210
Description	Physical Form	Storage Temperature
The antibody was produced from stable CHO clone 3-B9 (July 2010) in 1 x 5 L bioreactor run at Lpath.	PBS (Cellgro, # 21-040 Lot 21040190)	4 °C
Production Date	Retest Date	Project /Use
July 2010		TBD

Analytical Data		
Attribute	Method	Result
Concentration	Human IgG quantification ELISA	15.3 mg/ml
Potency (EC50)	S1P/LPA Direct binding ELISA	Does not bind
Molecular weight	SDS-PAGE	135 kDa
Endotoxin assay	Cambrex endotoxin assay kit	0.02 EU/mg protein
Structural Integrity	SDS-PAGE and SEC-HPLC	See attached data

TITLE	LT1023, human NS IgG antibody (15.3 mg/ml)				LOT NUMBER
					LP101210
DOCUMENT NUMBER	REVISION	SUPERSEDED REVISION	ISSUE DATE	EFFECTIVE DATE	PAGE
AB-COA-007	0.0.0	0.0.0	MAY 23, 2008	MAY 23, 2008	2 of 3

Aggregation	SEC-HPLC	# N/A (tailing also seen in reference standard)
S1P/LPA carry-over (Molar ratio of lipid to antibody)	S1P/LPA-Competitive ELISA	Below the level of quantification
Thermostability	Human IgG binding ELISA	Not done
Sterility	0.2 um filtration	Yes

Comments

Production location, date, and method

The antibody was produced from stable CHO clone 3-B9 (July 2010) in 1 x 5 L bioreactor run at Lpath.

Purification location, date and method

The antibody was purified at Lpath using affinity (MabSelect protein A) and AEX (Q membrane Sartorius Stedim 92IEXQ42D4-00-A) chromatography on July 30, 2010. The purified antibody was dialyzed against 3x 1.8 L PBS (Cellgro, # 21-040 Lot 21040190) and passed through a 0.2 uM bottle-top filter unit.

Concentration by A_{280} and extinction coefficient

The final sample is 15.3 mg/ml as determined by measuring the UV absorbance (280 nm) and extinction coefficient of 1.4 ml/mg.

Total amount of Ab (mg)

1.1 grams

Other comments on quality:

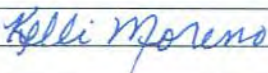
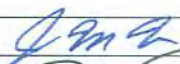
Solution is clear and colorless

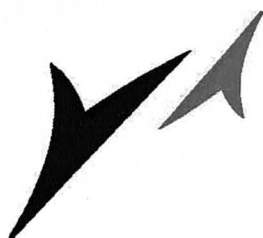
Ab showed tailing on HPLC, so aggregation quantification was not possible. This was also seen on the reference standard and seems to be an artifact of the HPLC run.

Showed no activity on IL-6 or IL-8 bio-assay.

Approval

		Certificate of Analysis			
TITLE	LT1023, human NS IgG antibody (15.3 mg/ml)				LOT NUMBER
					LP101210
DOCUMENT NUMBER	REVISION	SUPERSEDED REVISION	ISSUE DATE	EFFECTIVE DATE	PAGE
AB-COA-007	0.0.0	0.0.0	MAY 23, 2008	MAY 23, 2008	3 of 3

Prepared by (QC)	Kelli Moreno 	Date: <u>8/12/10</u>
		Date:
		Date:
Review by (R &D)	Jon Wojciak, Ph.D. 	Date: <u>8/12/10</u>
Approved by	Mary-Ann Campbell Ph.D. 	Date: <u>8/12/10</u>
		Date:



H14175 Received 10/14/14 RA

H14173 conjugated to 5-(6)-Carboxyfluorescein

Product Number	FAM-999
Lot Number	FAM999-14-006
Protein	LT1023, human NS IgG, Lot LP101210 (On CofA) Lot LP121424 (On Vial)
5(6)-FAM SE	Biotium 90024, Lot 7F0730
Volume	2.02 ml
Conc. (mg)	2.90 mg/mL
Amount	5.9 mg
F:P	2.0
Store at	2-8°C

Form/Shipping & Storage

Supplied as liquid. Upon receipt, store at 2-8°C.

Handling

We recommend that the investigator determine the appropriate working concentration for their specific application. Avoid exposure to heat and light.

Buffer

The product is provided in PBS, pH 7.4



COLUMBIA BIOSCIENCES

4985 Winchester Blvd. Frederick, MD 21703 p: 310.732.5415 f: 443.558.3883 www.columbiabiosciences.com

Appendix E. Quality Assurance Statement



QUALITY ASSURANCE STATEMENT

Client Number: AZ-366

Study Number: DGG14-010

Study Title: Full Human Tissue Cross-Reactivity Study of LT3114

This histopathology project has been inspected and audited by the Alizée Pathology Quality Assurance Unit (QAU) as required by the Good Laboratory Practice (GLP) regulations disseminated by the U.S. Food and Drug Administration.

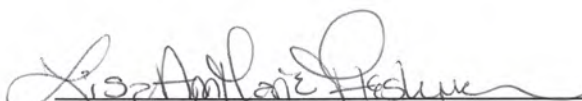
Area Inspected	Dates		
	<u>Inspection¹</u>	<u>Reported²</u>	<u>Reported³</u>
Critical Phase: Dilutions	4Dec14	5Jan15	5Jan15
Data Review	29-31Dec14	13Jan15	13Jan15
Draft Report	29-31Dec14;2Jan15	5Jan15	5Jan15
Final Report	13Jan15	13Jan15	13Jan15

All the results/conclusions of the tissue cross-reactivity report accurately reflect the raw data. The most updated version of the company organizational chart dated 18Nov14 was used to verify that the Alizée Pathology QAU is operationally independent from study managers.

¹ Date(s) of inspection

² Date(s) inspections reported to Alizée Pathology Management, Principal Investigator/Study Pathologist

³ Date(s) inspections reported to Study Director, Test Facility Management, and Lead QA (if appropriate)


AnnMarie Fleshman
Quality Assurance Auditor

13 January 2015
Date